

BIOL 250 — Pelvis and Perineum

Comprehensive Study Notes

Ch 27 — The Reproductive System

27.1 | Organization of the Reproductive System

- The reproductive system consists of: **gonads** (produce gametes and hormones), a **reproductive tract** of ducts that receive/store/transport gametes, **accessory glands and organs** that secrete fluids into the ducts, and **external genitalia** (perineal structures).
- **Fertilization (conception)**: fusion of sperm (father) and oocyte (mother); produces a **zygote**, a single diploid cell. Development to a full-term infant takes ~9 months.
- Male gonads = **testes**: secrete **androgens** (primarily **testosterone**) and produce ~half a billion sperm per day. Sperm travel a duct system, mix with accessory gland secretions to form **semen**, and are expelled during **ejaculation**.
- Female gonads = **ovaries**: typically produce only one immature gamete (**oocyte**) per month. Oocyte travels the **uterine tubes** (Fallopian tubes) into the muscular **uterus**; the **vagina** connects uterus to exterior. During intercourse, semen is introduced into the vagina and sperm ascend the tract to encounter an oocyte.
- **Descent of the Testes**: Testes form in the body cavity adjacent to the kidneys during development. The **gubernaculum testis** (cord of connective tissue/muscle) extends from the inferior testis to the posterior wall of a peritoneal pocket; as the fetus grows the gubernaculum does not elongate, holding testes in place while the body grows around them. In the 7th developmental month, hormonally stimulated contraction of the gubernaculum pulls the testes through the **inguinal canal** to the scrotum (**descent of the testes**). Small peritoneal pockets accompany this migration.
- **Spermatic Cords**: paired structures from the abdominopelvic cavity to the testes. Begin at the **deep inguinal ring**, pass through the **inguinal canal**, exit at the **superficial inguinal ring**, descend to the scrotum. Contain: **ductus deferens**, **deferential artery**, **testicular artery**, **pampiniform plexus** (of testicular vein), **ilio-inguinal** and **genitofemoral nerves** (from lumbar plexus), lymphatics — all wrapped in fascia/muscle layers.
- **Inguinal canals**: narrow canals linking scrotal chambers to the peritoneal cavity; usually close after descent, but the spermatic cord's presence creates permanent weak points in the abdominal wall → **inguinal hernias are common in males**. Inguinal hernias in women are rare (their inguinal canals are much smaller, no spermatic cord).
- Concept check emphasis: gonads produce gametes AND hormones.

27.2 | Anatomy of the Male Reproductive System

Overview: sperm travel epididymis → ductus deferens → ejaculatory duct → urethra. Accessory organs (seminal glands, prostate, bulbo-urethral glands) secrete into ejaculatory ducts/urethra. External genitalia = scrotum (encloses testes) + penis (erectile organ carrying distal urethra).

The Testes

- Each testis: ~5 cm (2 in.) long, 3 cm (1.2 in.) wide, 2.5 cm (1 in.) thick; weighs 10–15 g (0.35–0.53 oz).
- Hang within the **scrotum**, a pouch of skin inferior to the perineum, anterior to the anus.
- **Scrotum**: divided internally into two chambers by the **raphe of the scrotum** (external ridge). Each testis occupies its own **scrotal cavity**.
- **Tunica vaginalis**: serous membrane lining the scrotal cavity; reduces friction between parietal and visceral layers.

- Scrotal wall layers (superficial to deep): skin → superficial fascia → **dartos muscle** (smooth muscle; resting tone elevates testes, wrinkles scrotal skin) → **cremaster muscle** (skeletal muscle; contraction via **cremasteric reflex** tenses scrotum, pulls testes toward body — occurs with sexual arousal and cold).
- Normal sperm development requires temperature ~1.1°C (2°F) lower than core body temp; the cremaster adjusts testicular position to maintain this (relaxes/moves testes away from body when warm; contracts/pulls testes close when cold, e.g., cold water).
- Scrotal innervation: hypogastric plexus, ilio-inguinal nerves, genitofemoral nerves, pudendal nerves. Scrotal blood supply: internal pudendal arteries (branches of internal iliac), external pudendal arteries (branches of femoral), cremasteric branch of inferior epigastric arteries (branches of external iliac). Veins parallel arteries.
- **Tunica albuginea**: dense fibrous connective tissue capsule surrounding testis, covered by tunica vaginalis; rich in collagen continuous with epididymal capsule. Extends inward as **septa testis**, converging toward the **mediastinum of the testis** (supports blood vessels, lymphatics, ducts).
- **Histology/lobules**: septa testis divide the testis into lobules containing ~800 slender, tightly coiled **seminiferous tubules** (each ~80 cm/31 in. long; total ~half a mile per testis) — site of sperm production.
- Each seminiferous tubule is U-shaped, connects to a **straight tubule** → **rete testis** (maze of interconnected passageways in the mediastinum) → **15–20 efferent ductules** → **epididymis**.
- Between tubules: numerous blood vessels and **interstitial endocrine cells (Leydig cells)**, which produce **androgens** (dominant male sex hormone = **testosterone**). Testosterone functions: stimulate spermatogenesis; promote physical/functional maturation of sperm; maintain accessory organs of male reproductive tract; cause development of secondary sexual characteristics (e.g., facial hair); stimulate growth/metabolism throughout the body; influence brain development (sexual behaviors/drive).

Spermatogenesis and Meiosis

- **Spermatogenesis** begins at the outermost layer of cells in seminiferous tubules with stem cells called **spermatogonia** (form during embryonic development, dormant until puberty).
- Beginning at puberty, spermatogonia divide continuously; one daughter cell remains an undifferentiated stem cell, the other becomes a **primary spermatocyte** that begins **meiosis**.
- **Meiosis**: cell division producing gametes with half the chromosome number (haploid, 23 individual chromosomes) vs. **mitosis** (two identical diploid daughter cells, 23 pairs).
- Primary spermatocyte → (meiosis I) → 2 **secondary spermatocytes** → (meiosis II) → 4 **spermatids** total per primary spermatocyte.
- Spermatogenesis is directly stimulated by testosterone and indirectly stimulated by **FSH** (follicle-stimulating hormone). Interstitial endocrine cells produce testosterone in response to **LH** (luteinizing hormone).

Spermiogenesis

- Spermatids differentiate into physically mature sperm while embedded in the cytoplasm of **nurse (Sertoli) cells**.
- Detachment of sperm from the nurse cell releases it into the tubule lumen — marks the end of spermiogenesis.
- Entire process (spermatogonia → mature sperm) takes **~9 weeks**.

Nurse (Sertoli) Cells — 5 functions

1. **Maintain the blood testis barrier** (comparable to blood-brain barrier): tight junctions between nurse cells isolate the tubule lumen from interstitial fluid/general circulation; tightly regulated transport maintains a stable luminal environment (fluid high in androgens, estrogens, potassium, amino acids — very different from interstitial fluid). Sperm carry sperm-specific antigens not found in somatic cells, which the immune system would attack without this barrier.
2. **Support mitosis and meiosis**: FSH and testosterone stimulate nurse cells, which promote spermatogonial division and meiotic divisions of spermatocytes.
3. **Support spermiogenesis**: nurse cells surround/enfold spermatids, providing nutrients and chemical stimuli.
4. **Secrete inhibin**: depresses pituitary FSH and hypothalamic GnRH production — negative feedback control of spermatogenesis.

5. **Secrete androgen-binding protein (ABP)**: binds androgens in tubule fluid, elevating intratubular androgen concentration, stimulating spermiogenesis; FSH stimulates ABP production.

Anatomy of a Sperm — three regions:

1. **Head**: densely packed chromosomes; tip contains the **acrosome** (membrane-bound vesicle with enzymes for fertilization).
2. **Neck**: short, attaches head to middle piece; contains both centrioles of the original spermatid; distal centriole connects to microtubules continuous through middle piece and tail; mitochondria spiral around the microtubules (provide ATP for tail movement).
3. **Tail**: only flagellum in the human body; whip-like, corkscrew motion (unlike ciliary beating).
 - Mature sperm lack ER, Golgi, lysosomes, peroxisomes, and most other organelles; lack glycogen/energy reserves — must absorb nutrients from surrounding fluid.

The Male Reproductive Tract

- ***Epididymis***: start of the reproductive tract, lies along the posterior testis; firm, palpable through the scrotum. Each epididymis is a coiled tubule ~7 m (23 ft) long, subdivided into **head, body, tail**.
 - Head receives sperm via efferent ducts from the mediastinum of the testis.
 - Body: begins distal to last efferent duct, extends along posterior testis margin; convolutions decrease near the inferior border marking the start of the tail.
 - Tail: reverses direction; epithelium changes from stereociliated pseudostratified columnar (head/body) to resemble that of the ductus deferens; tail is the main sperm storage site.
 - Functions: (1) monitors/adjusts fluid composition from seminiferous tubules — lining has **stereocilia** increasing surface area for absorption/secretion; (2) recycles damaged sperm/cellular debris via enzymatic breakdown, released into interstitial fluid for vascular pickup; (3) stores spermatozoa and facilitates functional maturation — takes **~2 weeks** to pass through epididymis; sperm leaving are mature but immobile — must undergo **capacitation** to become motile and fertilization-capable: (a) become motile upon mixing with seminal gland secretions, (b) become capable of fertilization after exposure to conditions in the female tract (changes sperm plasma membrane). A secretion of the epididymis prevents premature capacitation while sperm remain in the male tract.
 - Rhythmic stereocilia movement + peristaltic smooth muscle contraction push sperm toward the ductus deferens.
 - **Clinical**: Testicular cancer usually affects young/middle-aged men, arises from rapidly dividing spermatogonia; often presents as a painless lump; best defense is regular testicular self-exam; treatment often includes **orchiectomy** (surgical testis removal) + chemotherapy; often curable if caught early.
- ***Ductus Deferens (vas deferens)***: 40–45 cm (16–18 in.) long; begins at the tail of the epididymis, ascends through the inguinal canal as part of the spermatic cord into the abdominopelvic cavity. Passes posteriorly/inferiorly along the lateral bladder surface, medial and anterior to the ureters, then descends along the posterior bladder surface toward the prostate. Enlarges near the prostate to form the **ampulla of the ductus deferens**. Wall has thick smooth muscle producing peristaltic contractions; lined by pseudostratified columnar epithelium with stereocilia. Also stores sperm for months in a low-metabolic inactive state. Each ampulla joins the excretory duct of the seminal gland on that side to start the **ejaculatory duct** (≤ 2 cm), which penetrates the prostate and empties into the prostatic urethra.
- ***Urethra***: extends from urinary bladder to tip of penis, 15–20 cm (6–8 in.); divided into **prostatic, membranous** (intermediate part), and **spongy** regions. Serves both urinary and reproductive systems in males.

The Accessory Glands — activate sperm, provide nutrients for motility, produce buffers against urethral/vaginal acidity.

- ***Seminal Glands (seminal vesicles)***: paired, embedded in connective tissue between posterior bladder wall and anterior rectal wall; each ~15 cm (6 in.) tubular gland with short side branches, coiled/folded into a compact mass ~5 x 2.5 cm (2 x 1 in.). Lined by pseudostratified columnar epithelium; highly active secretory glands contributing **~60% of semen volume**. Secretion: same osmotic concentration as blood plasma but different composition — contains **prostaglandins, clotting proteins, high fructose** (ATP substrate for sperm). Discharged into ductus deferens during ejaculation via sympathetically-controlled peristaltic contractions

(ductus deferens → seminal glands → prostate). Mixing with seminal fluid activates previously inactive sperm to beat flagella and become motile.

- ***Prostate***: small, muscular, rounded organ ~4 cm (1.6 in.) diameter; encircles the prostatic urethra. Contains **30–50 compound tubulo-alveolar glands**, lined mainly by simple columnar epithelium (varies simple squamous–cuboidal–pseudostratified columnar), wrapped in a thick blanket of smooth muscle. Produces **prostatic fluid**, weakly acidic, **20–30% of semen volume**; rich in enzymes that nourish sperm and prevent coagulation in the vagina; contains **seminalplasmin**, an antibiotic that may help prevent UTIs in males. Peristaltic contractions eject secretions into the prostatic urethra.
- ***Bulbo-urethral glands (Cowper's glands)***: paired, located at the base of the penis; small round glands ~10 mm diameter. Duct of each parallels the spongy urethra 3–4 cm (1.2–1.6 in.) before emptying into the urethral lumen. Lined by simple cuboidal–columnar epithelium; compound tubulo-alveolar mucous glands secreting thick, sticky, alkaline mucus — neutralizes residual urinary acids in the urethra and lubricates the glans/tip of penis. Contribute ~5% of semen volume.

The Penis

- Tubular organ for intercourse and conducting urine; regions: **root** (attaches penis to ischial rami within the urogenital triangle, inferior to pubic symphysis), **body/shaft** (tubular, movable, contains erectile tissue), **glans penis** (expanded distal end surrounding the external urethral orifice).
- Skin: hairless, more pigmented; dermis has smooth muscle; loose underlying connective tissue allows movement; subcutaneous layer has superficial vessels/lymphatics but few fat cells.
- **Foreskin (prepuce)**: fold of skin covering the glans, attaches at the neck of the glans. **Preputial glands** in the neck/inner foreskin secrete **smegma** (waxy material; excellent bacterial nutrient source → mild inflammation/infection common if not washed). **Circumcision** = surgical removal of foreskin, a preventive measure.
- Erectile tissue: three parallel cylindrical columns — a maze of vascular channels separated by elastic connective tissue/smooth muscle partitions; encircled by a dense elastic fiber layer.
 - **Corpora cavernosa** (paired): on the posterior/dorsal surface, encircled by dense collagenous sheath, separated by a thin septum; diverge at their bases forming the **crus of the penis**, each bound to the ischial ramus/pubis by connective tissue ligaments; extend to the neck of the penis; surround a central **deep artery of the penis**.
 - **Corpus spongiosum**: surrounds the spongy urethra; extends from superficial fascia to the tip, expanding to form the **glans penis**; sheath has more elastic fibers than corpora cavernosa; contains a pair of small arteries; the **bulb of the penis** is its thickened proximal end.
 - Resting state: arterial branches constricted, muscular partitions tense → restricted blood inflow. Erection: smooth muscle relaxes, erectile tissue engorges with blood, penis stiffens/elevates.
- **Ejaculation (two-step process)**:

1. **Emission**: sympathetic nervous system coordinates peristaltic contractions sweeping from ductus deferens → seminal glands → prostate → bulbo-urethral glands, mixing semen components.
2. **Ejaculation**: powerful rhythmic contractions of **ischiocavernosus** (inserts along penis sides, stiffens organ) and **bulbospongiosus** (wraps around penis base, pushes semen toward external urethral orifice) — two pelvic floor muscles. Controlled by reflexes involving inferior lumbar and superior sacral spinal cord segments.

Semen

- Typical ejaculation releases **2–5 mL** of semen.
- Components:

1. **Sperm**: normal count **20–100 million sperm/mL**.
2. **Seminal fluid**: mixture of glandular secretions — seminal glands (60%), prostate (30%), nurse cells/epididymis (5%), bulbo-urethral glands (5%).
3. **Enzymes**: (a) a protease to help dissolve vaginal mucus; (b) **seminalplasmin** — antibiotic, kills bacteria incl. *E. coli**; (c) a prostatic enzyme converting fibrinogen → fibrin, coagulating semen within minutes of ejaculation; (d) **fibrinolysin**, liquefies clotted semen after 15–30 minutes.

Clinical Case (Testicular Torsion): Liam's twisted spermatic cord compresses the testicular artery (inflow) and pampiniform plexus (outflow), and twists the ilio-inguinal/genitofemoral nerves. Signs: tender, elevated/malpositioned testis, epididymis displaced anteromedially, absent cremasteric reflex on affected side (stroking inner thigh normally retracts testis toward inguinal canal via the reflex), swelling. Diagnosed via bedside ultrasonography showing absent blood flow; treated with emergency surgical detorsion. Epididymitis can mimic torsion but cremasteric reflex typically remains intact.

27.3 | Anatomy of the Female Reproductive System

Overview: Oocytes form in ovaries → travel uterine tubes (fertilization site) → reach uterus → uterus opens to vagina → external genitalia surround vaginal opening. Accessory glands secrete into the tract as in males.

Supporting structures

- **Broad ligament:** large mesentery enclosing ovaries, uterine tubes, and uterus; attaches to the sides/floor of the pelvic cavity, continuous with parietal peritoneum; limits side-to-side movement/rotation of the uterus.
 - **Mesosalpinx:** free edge of the broad ligament attached to each uterine tube.
 - **Mesovarium:** thickened fold of broad ligament supporting/stabilizing each ovary's position.
 - Subdivides the pelvic cavity into: **recto-uterine pouch** (between posterior uterus and anterior colon) and **vesico-uterine pouch** (between anterior uterus and posterior bladder).
- Other ligaments (within the broad ligament sheet) prevent superior-inferior movement — see Uterus section below.

The Ovaries

- Flattened oval, ~5 cm long x 2.5 cm wide x 8 mm thick (2 x 1 x 0.33 in.); weigh 6–8 g (~0.25 oz). Produce oocytes and secrete hormones.
- No peritoneal covering; stabilized by the **mesovarium**, **ovarian ligament** (extends from lateral uterine wall near the uterine tube attachment to the medial ovary surface), and **suspensory ligament** (extends from the lateral ovary surface past the uterine tube's open end to the pelvic wall; contains the **ovarian artery** and **ovarian vein**, which attach at the **ovarian hilum**).
- Pink/yellowish, nodular. Covered by a single layer of cuboidal **germinal epithelium** overlying a dense connective tissue **tunica albuginea**. Interior stroma = superficial **cortex** (site of gamete production) + deeper **medulla**.
- **Clinical — Ovarian Cancer:** uncommon but deadly; usually arises in the germinal epithelium; no screening exam exists; asymptomatic early, often undetected until spread within the peritoneum; advanced disease treated with surgery + chemo. Symptoms of advanced disease: lower abdominal pain/heaviness, abnormal vaginal bleeding, abnormal periods, weight gain/loss, gas/nausea/appetite loss. Full-term pregnancy before age 26 lowers risk (risk decreases with each full-term pregnancy); oral contraceptives, tubal ligation, hysterectomy (ovaries retained) also reduce risk. Risk increases with age and family history; **BRCA1/BRCA2** mutations linked to ovarian and breast cancer — BRCA1 is a tumor-suppressor gene; normal-BRCA1 lifetime ovarian cancer risk is <2%.

The Ovarian Cycle and Oogenesis

- Three interrelated processes: **oogenesis** (production of oocytes), the **ovarian cycle** (monthly maturation of an oocyte), the **uterine cycle** (prepares uterus for implantation).
- Oogenesis begins before birth, accelerates at puberty, ends at menopause.
- During fetal development, **oogonia** (female reproductive stem cells) produce diploid **primary oocytes (2n)**. Ovarian follicles (in the cortex) are sites of oocyte growth/oogenesis.
- At puberty, rising **FSH** triggers the ovarian cycle; monthly, FSH stimulates some primordial follicles to develop further.
- **Stages of follicle development:**

1. **Primordial ovarian follicles:** primary oocytes located in outer ovarian cortex near the tunica albuginea, in clusters called **egg nests**; each surrounded by a single squamous layer of follicle cells.

2. **Primary ovarian follicles:** follicular cells become cuboidal, divide repeatedly forming multiple layers = **granulosa cells**; oocyte secretes proteins forming the **zona pellucida** (extracellular coat); granulosa cell microvilli + oocyte microvilli penetrate the zona pellucida for nutrient transfer. Adjacent stromal cells form a layer of **thecal endocrine cells** around the follicle. Thecal + granulosa cells together produce **estrogens**; **estradiol** is the most important/dominant estrogen prior to ovulation. Estrogen functions: stimulate bone/muscle growth; maintain female secondary sex characteristics; affect CNS activity (sex behaviors/drives); maintain reproductive gland/organ function; initiate repair/growth of the uterine lining.

3. **Secondary ovarian follicles:** only a few primary follicles mature further (regulated by an oocyte-produced growth factor); follicle wall thickens.

4. **Tertiary (vesicular) follicle:** over 2–3 months, follicular fluid accumulates in pockets between follicle cell layers, doubling follicle size; large size bulges the ovary surface; oocyte + surrounding follicle cells project into the fluid-filled **antrum**; granulosa cells immediately around the oocyte form the **corona radiata**. Primary oocyte (suspended in prophase I) completes the first meiotic division here, producing a **secondary oocyte** + a small nonfunctional **first polar body** (unequal cytoplasm division — oogenesis differs from spermatogenesis in this respect). The secondary oocyte proceeds to metaphase of meiosis II but arrests there unless fertilization occurs; if fertilized, meiosis II completes, producing an **ovum** + a **second polar body**. (Net result: oogenesis yields one functional ovum + polar bodies, vs. spermatogenesis's 4 equal gametes.)

5. **Ovulation:** the tertiary follicle releases the secondary oocyte; the follicular wall ruptures, releasing contents (incl. secondary oocyte) into the pelvic cavity; sticky follicular fluid keeps the corona radiata attached near the ruptured follicle wall; oocyte is moved into the uterine tube by contact with **fimbriae** or ciliary currents. Normally only one oocyte released per cycle, though multiple follicles may have partially developed (undergo **follicular atresia**).

6. **Corpus luteum formation:** the empty tertiary follicle collapses, ruptured vessels bleed; under **LH** stimulation, remaining granulosa cells proliferate to form the **corpus luteum**, which produces **progestins** (mainly **progesterone**).

7. **Corpus albicans formation:** without fertilization, the corpus luteum degenerates ~**12 days after ovulation**; progesterone/estrogen levels decline; fibroblasts invade forming a scar-tissue knot, the **corpus albicans** — marks the end of the ovarian cycle.

- Declining progesterone/estrogen at cycle's end stimulates hypothalamic **GnRH**, triggering a rise in FSH/LH from the anterior pituitary, starting a new cycle.

The Uterine Tubes

- Hollow muscular tubes, ~13 cm (5 in.) long each; five regions:

1. **Fimbriae:** numerous fingerlike projections draping over the ovary surface.

2. **Infundibulum:** expanded funnel closest to the ovary; ciliated lining beats toward the tube's middle.

3. **Ampulla:** middle region; site where **fertilization typically occurs**.

4. **Isthmus:** short region connecting ampulla to the uterine wall.

5. **Uterine part:** continuation of the isthmus, opens into the uterine cavity.

- Histology: epithelium has both ciliated and nonciliated simple columnar cells; concentric smooth muscle layers surround the mucosa. Transport = ciliary movement + peristalsis. A few hours before ovulation, sympathetic/parasympathetic fibers from the hypogastric plexus "turn on" ciliary beating.
- Oocyte transit time infundibulum → uterine cavity: **3–4 days**. Fertilization must occur within the **first 12–24 hours** of the oocyte's passage.
- Tube provides a nutritive lipid/glycogen-rich environment for sperm and developing pre-embryo. Unfertilized oocytes degenerate in the terminal uterine tube or uterus.
- **Uterine tube histology detail:** isthmus and ciliated epithelial surface shown via LM/SEM images.
- **Retroflexion** (uterus bends backward toward sacrum) occurs in ~**20% of adult women**, no clinical significance (contrast with normal **anteflexion**).

The Uterus

- Empty uterus: small pear-shaped organ, ~7.5 cm (3 in.) long, max diameter 5 cm (2 in.); weighs **30–40 g** (1–1.4 oz).
- Normally bends anteriorly near its base = **anteflexion** (uterine body lies across the posterior/superior bladder surface). Backward bending toward sacrum = **retroflexion** (~20% of women, no clinical significance).
- Gross regions: **body** (largest region) → **fundus** (rounded portion superior to uterine tube attachment) → constriction at **isthmus of uterus** → **cervix** (inferior portion, isthmus to vagina) → projects ~1.25 cm (0.5 in.) into the vagina, forming a curved surface around the **external os** (external opening) → leads to the **cervical canal** → **internal os** (internal opening) → **uterine cavity**.
- Cervical mucus: viscous, fills the cervical canal, covers external os, prevents bacterial passage from vagina; becomes watery as ovulation approaches (viscous mucus impairs sperm entry/fertility — infertility treatment may include drugs to thin cervical mucus).
- **Suspensory ligaments of the uterus** (3 pairs, plus the broad ligament):
 - **Uterosacral ligaments**: lateral uterus → anterior sacrum; prevent inferior/anterior movement.
 - **Round ligaments**: lateral uterus (inferior to uterine tube bases) → anteriorly through the inguinal canal → connective tissue of external genitalia; restrict posterior movement.
 - **Cardinal (transverse cervical) ligaments**: base of uterus/vagina → lateral pelvic walls; prevent inferior movement.
 - Pelvic floor skeletal muscles/fascia provide additional support.
- **Uterine wall layers**: outer **perimetrium** (incomplete serosal layer, continuous with peritoneum, covers fundus and anterior/posterior body surfaces) → **myometrium** (outer, muscular; thickest wall layer; 3 poorly-defined smooth muscle layers — middle layer thickest, circular/spiral; inner & outer layers longitudinal; provides force to expel the fetus) → **endometrium** (inner, glandular).
 - Endometrium subdivided into: **functional layer** (closest to uterine cavity, contains most uterine glands, contributes most endometrial thickness, undergoes cyclic change) and **basal layer** (attaches endometrium to myometrium, contains terminal gland branches, structurally constant, survives menstruation).
 - Uterine glands extend from the endometrial surface deep toward the myometrium; under estrogen influence, glands/vessels/endothelium change through the uterine cycle to support a growing fetus.
- **Blood supply**: **uterine arteries** (branches of internal iliac arteries) and **ovarian arteries** (branches of abdominal aorta inferior to the renal arteries), with extensive interconnections ensuring reliable flow despite positional/shape changes in pregnancy. Within the myometrium: **arcuate arteries** encircle the endometrium → **radial arteries** branch off → supply **straight arteries** (to basal layer) and **spiral arteries** (to functional layer).
- **Innervation**: autonomic fibers from the hypogastric plexus (sympathetic) and sacral segments **S3–S4** (parasympathetic); sensory afferents enter the spinal cord via dorsal roots of **T11–T12**.
- **Clinical**: Uterine fibroids — most common benign uterine tumor, arise in the myometrium, not cancerous; may impair fertility, cause heavy/painful periods or intermenstrual bleeding. Cervical cancer — caused by certain **HPV** strains (sexually transmitted); early disease usually asymptomatic; detectable via pelvic exam + **Pap smear** (named for Dr. George Papanicolaou); HPV vaccination could theoretically eliminate many cases; pre-cancerous lesions/early cancers have good prognosis. Endometrial cancer — most common uterine cancer type, arises in the endometrium; most cases age 50–70; estrogen replacement therapy without progesterone increases risk; post-menopausal bleeding is the most common sign and should always be investigated; Pap smear may not detect it; good prognosis if treated via hysterectomy ± radiation before spread.

The Uterine Cycle (Menstrual Cycle)

- Averages **28 days** (range 21–35 days in normal individuals); three phases: **menstrual, proliferative, secretory**, driven by hormones regulating the ovarian cycle.
- **Menstrual phase**: onset marked by degeneration of the endometrial functional layer. Arteries nourishing the functional layer constrict, weakening arterial walls/reducing flow; secretory glands/tissues of the functional layer die; weakened arterial walls rupture, blood pours into connective tissue; blood cells and degenerating tissue break away and shed via the external os into the vagina = **menstruation (menses)**, lasting **1–7 days**; blood loss ~**35–50 mL** (1.2–1.7 oz). **Dysmenorrhea** (painful menstruation) may result from uterine inflammation/contraction or adjacent pelvic pathology. Triggered by declining progesterone/estrogen at the end of the ovarian cycle; continues until the next follicle group raises estrogen again.

- **Proliferative phase:** basal layer circulation (incl. basal gland portions) remains constant during menstruation so the basal layer survives. Post-menstruation, rising circulating estrogens stimulate epithelial gland cells to multiply and spread across the endometrial surface, restoring epithelial integrity, growth, and vascularization of the functional layer. Occurs concurrently with primary/secondary follicle enlargement in the ovary; sustained by estrogens from developing follicles. By ovulation, the functional layer is several mm thick, highly vascularized; mucous glands secrete glycogen-rich mucus essential for early survival of a fertilized ovum.
- **Secretory phase:** begins at ovulation, persists as long as the corpus luteum remains intact (~14 days); endometrial glands enlarge, accelerate secretion; endometrial arteries elongate and spiral through the functional layer; driven by combined progestin + estrogen stimulation from the corpus luteum. Secretory activity peaks ~12 days after ovulation; glands become less active over the next 1–2 days as the corpus luteum stops producing hormones, ending the cycle → menstruation begins again. Secretory phase usually lasts 14 days, so ovulation date can be back-calculated as 14 days before the next menstrual onset.
- **Menarche:** first uterine cycle, usually age **11–12**. **Menopause:** last uterine cycle, age **45–55**. Cycles interrupted only by illness, stress, starvation, pregnancy. Irregular cycles are typical for the first 2 years after menarche and last 2 years before menopause.

The Vagina

- Average length **7.5–9 cm** (3–3.5 in.); highly distensible so length/width vary. Proximal end: cervix projects into the vaginal canal; **vaginal fornix** = shallow recess around the cervical protrusion. Lies parallel and posteriorly in close contact with the rectum. Urethra travels along its superior wall anteriorly.
- Blood supply: **vaginal branches of the internal iliac arteries** (sometimes branching from the uterine arteries instead); venous drainage via the **vaginal vein**; innervation from the hypogastric plexus, sacral nerves S2–S4, and branches of the **pudendal nerve**.
- Three major functions: (1) passageway for menstrual fluid elimination; (2) receives the penis during intercourse, holds sperm before uterine passage; (3) forms the lower birth canal during childbirth.
- Walls moistened by cervical gland secretions and transudate across the permeable epithelium. Muscular layer + outer adventitial layer have extensive vascular/lymphatic networks.
- **Hymen:** elastic epithelial fold separating vagina and vestibule; may partially/completely block the vaginal entrance. **Bulbospongiosus muscles** flank the vaginal orifice; contraction constricts the entrance; these cover the **bulb of vestibule** (erectile tissue mass on each side of the vaginal entrance), embryologically homologous to the male corpus spongiosum.
- Histology: mucosal **stratified squamous epithelium**, forms folds (**rugae**) when relaxed; thick elastic lamina propria with small vessels, nerves, lymph nodes; inner circular + outer longitudinal smooth muscle; serosal covering near the uterus (continuous with pelvic peritoneum), elsewhere a highly elastic adventitia.
- Contains resident bacteria supported by cervical mucus nutrients; bacterial metabolism creates an acidic environment restricting pathogen growth but also inhibiting sperm motility — hence seminal fluid buffers are important for fertilization success.

The External Genitalia (Vulva/Pudendum)

- **Vulva (pudendum):** region enclosing the female external genitalia.
- Vagina opens into the **vestibule**, bounded by the **labia minora** (smooth, hairless skin).
- **Urethra** opens into the vestibule anterior to the vaginal entrance; **urethral glands** discharge near the external urethral orifice.
- **Clitoris:** anterior to the external urethral orifice, projects into the vestibule; contains erectile tissue homologous to the male corpora cavernosa; engorges with blood during arousal; small erectile **glans** at its tip; extensions of the labia minora form its **prepuce**.
- **Lesser vestibular glands:** variable number, discharge onto the vestibule surface, keeping it moist.
- **Greater vestibular glands:** paired, discharge via ducts into the vestibule near the posterolateral margins of the vaginal entrance during arousal; mucous glands, homologous to male bulbo-urethral glands.
- **Mons pubis:** anterior limit of the vulva; adipose tissue bulge anterior to the pubic symphysis.
- **Labia majora:** lateral limits of the vulva; homologous to the male scrotum; contain adipose tissue; encircle/partially conceal the labia minora and vestibular structures; outer surface has coarse hair (like the mons).

pubis), inner surface relatively hairless; sebaceous glands and apocrine sweat glands on the inner surface moisten/lubricate.

The Breasts (Mammary Glands)

- Specialized accessory organs of the female reproductive system; produce milk (**lactation**) to nourish a newborn, whose other systems are not yet fully developed at birth.
- Located in subcutaneous tissue of the pectoral fat pad, lateral to the sternum on the chest wall.
- **Nipple**: small conical projection where mammary ducts open onto the body surface. **Areola**: reddish-brown skin around the nipple; large sebaceous glands give it a granular texture.
- Glandular tissue (apocrine) organized into separate **lobes**, each containing several secretory **lobules**; ducts from lobules converge into a single **lactiferous duct** per lobe; near the nipple, each duct expands into a **lactiferous sinus** (chamber). **15–20 lactiferous sinuses** typically open onto each nipple surface.
- **Suspensory ligaments of the breast**: dense connective tissue bands originating in the dermis, surrounding the duct system, extending between lobes/lobules.
- A layer of loose connective tissue separates the breast from underlying pectoralis muscles. Blood supply: branches of the **internal thoracic artery**. (Lymphatic drainage detailed in Ch 23.)
- Resting (non-pregnant) breast: dominated by duct system, not active glandular cells; breast size reflects adipose tissue amount, not glandular tissue amount; secretory apparatus doesn't develop until pregnancy.
- **Clinical — Breast cancer**: 1 in 8 American women will develop breast cancer in her lifetime; only ~10% due to inherited mutations, but **BRCA1/BRCA2** mutations significantly raise lifetime risk; rare in men but possible. Risk-reduction: healthy weight, physical activity, limiting alcohol; screening via self-exam and mammography (annual mammograms recommended starting mid-40s). Most breast cancers begin in the lactiferous ducts (**ductal carcinoma**); some in the lobules (**lobular carcinoma**). Breasts of different sizes have similar numbers of ducts/lobules — size differences are due to adipose tissue. Staging based on tumor size, lymph node spread, distant metastases (lymphatic spread means draining lymph nodes are sampled). Tumors graded by growth/spread rate. Hormone receptor testing (estrogen/progesterone receptor status) guides hormone therapy eligibility. Signs: lump/thickening, bloody nipple discharge, change in breast size/shape/appearance, skin changes (dimpling, redness, flaking, pitting like orange peel — due to tumor involvement of the suspensory ligaments of the breast pulling on skin), newly inverted nipple.
- **Development during pregnancy**: requires anterior pituitary hormones **prolactin (PRL)** and **growth hormone (GH)**, plus **human placental lactogen (HPL)** from the placenta. Ducts become mitotically active, gland cells appear; by end of 6th month of pregnancy, mammary glands are fully developed and begin producing secretions stored in the duct system. Milk release (**let-down**) triggered by infant suckling → stimulates posterior pituitary to release **oxytocin** → oxytocin triggers contraction of smooth muscle in lactiferous duct/sinus walls, ejecting milk.

27.4 | Aging and the Reproductive System

- **Key Point**: female reproductive aging (menopause) is a striking, relatively abrupt change; male reproductive aging (climacteric) is gradual and occurs over a longer period.
- **Menopause**: time when ovulation and menstruation cease; typically occurs at age **45–55** (text also states "around age 50"). In the years preceding it, ovarian/menstrual cycles become irregular. **Premature menopause** occurs before age 40, underlying cause = shortage of primordial follicles.
- Menopause accompanied by a rapid, sustained rise in **GnRH, FSH, LH**, while circulating **estrogen and progesterone decline**. Effects of declining estrogen: (1) reduces uterine size; (2) reduces breast size; (3) thins urethral and vaginal walls; (4) weakens connective tissue support of ovaries, uterus, vagina. Also linked to **osteoporosis** (bone brittleness) and cardiovascular/neural effects including hot flashes, anxiety, depression.
- **Male climacteric**: gradual changes, circulating testosterone begins declining between ages **50–60**, coupled with rising FSH and LH. Sperm production continues (men can father children into their 80s), but there is a gradual reduction in sexual activity in older men, possibly linked to declining testosterone. Testosterone replacement therapy is sometimes suggested to enhance libido in older men (and women).
- Clinical correlation: prolapsed uterus (uterus moves to a more inferior position in the pelvic cavity) is more common in postmenopausal women due to weakened supportive connective tissue.

Related Clinical Terms (Ch 27 glossary)

- **Amenorrhea:** menstruation not starting before age 16, or cessation of menstruation for ≥ 6 months during reproductive age.
- **Cervical dysplasia:** abnormal growth of epithelial cells in the uterine cervix; may progress to cancer.
- **Endometriosis:** growth of endometrial tissue outside the uterus.
- **Genital herpes:** STD caused by a herpes virus; painful genital blisters.
- **Gonorrhea:** STD caused by *Neisseria gonorrhoeae* ("the clap").
- **Gynecology:** branch of medicine dealing with female reproductive function/disease.
- **Gynecomastia:** development of breast tissue in males; consequence of some chemotherapeutic agents and marijuana use (among other causes).
- **Hydrocele:** accumulation of serous fluid in a body sac, especially the tunica vaginalis of the testis or along the spermatic cord.
- **Menorrhagia:** extremely heavy menstrual bleeding.
- **Oophorectomy:** surgical removal of one or both ovaries.
- **Oophoritis:** inflammation of the ovaries.
- **Orchitis:** inflammation of the testis.
- **Polycystic ovary syndrome (PCOS):** most common endocrinopathy in women; most common cause of anovulatory infertility.
- **Premature ejaculation:** ejaculating sooner than desired during intercourse; ~30% of men regularly experience this.
- **Premenstrual dysphoric disorder (PMDD):** physical/emotional symptoms occurring 5–11 days before menstruation (>150 associated signs/symptoms reported).
- **Premenstrual syndrome (PMS):** physical/mood disturbances in the last half of the menstrual cycle after ovulation, ending with onset of menstrual flow.
- **Pseudohermaphrodites:** individuals with external genitalia that do not match their gonads.
- **Salpingitis:** inflammation of a uterine tube.
- **Uterine prolapse:** pelvic floor muscles/ligaments stretch and weaken, providing inadequate uterine support, so the uterus descends into the vaginal canal.
- **Vasectomy:** surgical removal of a segment of each ductus (vas) deferens, preventing sperm from reaching distal portions of the male reproductive tract (male sterilization).
- **Vulvovaginal candidiasis:** common vaginal yeast infection, usually caused by *Candida albicans*.

Ch 26 — The Urinary System (Bladder, Urethra, Ureters — Pelvic Portions)

26.1 | The Kidneys (tail end — collecting system/regulatory structures, carried over into pelvic reading range)

- **Distal convoluted tubule (DCT):** ascends out of the medulla into the cortex, passes between the afferent/efferent arterioles at the vascular pole. Differs from the proximal convoluted tubule (PCT) in 3 ways: (1) smaller diameter, (2) epithelial cells lack microvilli, (3) cell boundaries easily visible. Functional difference: PCT = reabsorption; DCT = secretion.
- DCT functions: (1) active secretion of ions, acids, other materials; (2) reabsorption of sodium and calcium ions; (3) reabsorption of water (concentrates tubular fluid). Sodium transport in the distal tubule is controlled by circulating **aldosterone** from the adrenal cortex.

- **Juxtaglomerular complex:** regulates blood pressure and filtrate formation; releases **renin** and stimulates **erythropoietin**, which elevate blood volume, hemoglobin, and blood pressure, restoring normal filtrate production. Three specialized cell types:

1. **Macula densa:** monitors Na⁺ concentration in tubular fluid, regulates glomerular filtration rate and renin release from juxtaglomerular cells.

2. **Juxtaglomerular cells:** modified smooth muscle cells in the afferent arteriole wall; secrete renin.

3. **Extraglomerular mesangial cells:** located between afferent/efferent arterioles; provide feedback control between macula densa and juxtaglomerular cells.

- **Collecting system:** DCT (last nephron segment) opens into connecting tubules → collecting ducts → papillary ducts → minor calyx. Epithelium: simple cuboidal (connecting tubules) → columnar (collecting/papillary ducts). Function: transports fluid from nephron to renal pelvis, makes final osmotic/volume adjustments. **ADH** controls collecting system permeability: low permeability → dilute urine (fluid flows to renal pelvis); high permeability → water flows osmotically into the medulla → small volume of concentrated urine. Higher circulating ADH = more water reabsorbed = more concentrated urine.
- **Clinical — Kidney disease/dialysis:** Kidneys filter all blood roughly every 30 minutes. End-stage kidney disease = loss of 90% of kidney function; nearly half a million Americans on dialysis; ~200,000 have a functioning transplant; >100,000 awaiting transplant, <20,000 transplants performed/year. **Acute renal failure:** sudden loss, often from trauma/MI/reduced renal blood supply; often reversible with temporary dialysis. **Chronic renal failure:** gradual loss, may be asymptomatic until significant; leading causes = diabetes and hypertension; other causes: genetic disease, infection, kidney stones, autoimmune disease, certain drugs. Management starts with diet restricting fluids, protein, sodium, potassium, phosphorus. **Dialysis:** removes wastes/salt/water, balances electrolytes/pH, but cannot produce renin, erythropoietin, or calcitriol; performed ~3x/week via forearm shunt (or peritoneal dialysis using the peritoneal cavity as the dialysis membrane, manual or automated/overnight). **Kidney transplant** is the best treatment; donor kidney/ureter placed extraperitoneally in the pelvic cavity (iliac fossa), ureter connected to recipient's bladder; 2-year survival >90%, better than dialysis outcomes.

26.2 | Structures for Urine Transport, Storage, and Elimination

- Filtrate modification/urine production ends when fluid enters the minor calyx. Remaining urinary structures (**ureters, urinary bladder, urethra**) transport, store, and eliminate urine.
- **Transitional epithelium** lines the minor/major calyces, renal pelvis, ureters, urinary bladder, and proximal urethra — tolerates repeated distension/relaxation cycles without damage.

The Ureters

- Paired retroperitoneal muscular tubes, ~30 cm (12 in.) long, extending from the kidneys to the urinary bladder; begin as a funnel-shaped continuation of the renal pelvis. Pass inferiorly/medially over the psoas major. Anatomical course differs between sexes due to differences in reproductive organ size/position.
- Ureters penetrate the posterior bladder wall obliquely; **ureteral orifices are slit-like (not rounded)** — this shape prevents backflow of urine toward the kidneys when the bladder contracts.
- **Histology (3 layers):** (1) inner mucosa lined by transitional epithelium; (2) middle muscular layer (inner longitudinal + outer circular smooth muscle); (3) outer adventitia, continuous with the kidney's fibrous capsule and the parietal peritoneum.
- Peristaltic contractions triggered roughly every 30 seconds by stretch-receptor stimulation in the ureteral wall; contractions travel renal pelvis → bladder, "milking" urine through the ureter.

The Urinary Bladder

- Hollow muscular organ for temporary urine storage. In males, its base lies between the rectum and pubic symphysis; in females, inferior to the uterus and anterior to the vagina. A full bladder can hold about **1 liter** of urine.
- Peritoneum covers the superior surface when empty; as the bladder fills it becomes intraperitoneal, displacing parietal peritoneum anteriorly. Stabilizing peritoneal folds: **median umbilical ligament** (anterosuperior border of bladder → umbilicus) and **lateral umbilical ligaments** (sides of bladder → umbilicus) — these are fibrous

remnants of the two fetal **umbilical arteries** that supplied the placenta. Additional tough connective tissue bands anchor the posterior, inferior, and anterior bladder surfaces to the pelvic bones.

- Mucosa has folds (**rugae**) that flatten as the bladder fills/stretch.
- **Trigone**: smooth, thick triangular mucosal region at the base of the bladder; acts as a funnel channeling urine into the urethra when the bladder contracts. (Internal bladder features per chapter summary: trigone, neck, internal urethral sphincter.)
- **Neck of the urinary bladder**: region surrounding the urethral opening (most inferior point of the bladder); contains the muscular **internal urethral sphincter**, providing involuntary control via branches of the pelvic nerves (autonomic).
- **Histology**: mucosa (transitional epithelium with prominent rugae) → submucosa (connective tissue) → muscular layer of 3 smooth muscle layers (inner longitudinal, middle circular, outer longitudinal) collectively forming the **detrusor muscle** — contraction compresses the bladder and expels urine into the urethra → outer adventitia.

The Urethra

- Extends from the bladder neck to the exterior; male and female urethrae differ in length and function.
- **Female urethra**: very short, **3–5 cm (1–2 in.)**, from bladder to vestibule; **external urethral orifice** situated near the anterior vaginal wall.
- **Male urethra**: extends from bladder neck to the tip of the penis, **18–20 cm (7–8 in.)**; three subdivisions: (1) **prostatic urethra** (passes through the center of the prostate), (2) **membranous urethra** (intermediate part; short segment penetrating the muscular pelvic floor), (3) **spongy (penile) urethra** (from membranous urethra to the external urethral orifice at the penile tip).
- **External urethral sphincter**: skeletal muscle band in both sexes, acts as a valve; has resting muscle tone, must be voluntarily relaxed to permit urination; only the external sphincter is under **voluntary control** (via the perineal branch of the **pudendal nerve**). Both the internal and external urethral sphincters are influenced by branches of the **hypogastric plexus** (though only the external one is voluntary — its autonomic innervation matters mainly when voluntary control is lost, e.g., infants or after spinal cord injury).
- **Histology — female urethra**: transitional epithelium near the bladder neck; stratified squamous epithelium lines the rest. Lamina propria has an extensive venous network; concentric smooth muscle layers surround the whole complex.
- **Histology — male urethra**: epithelium changes along its length — transitional (near bladder) → pseudostratified/stratified columnar → stratified squamous (near external urethral orifice). Lamina propria thick/elastic; mucosa folded into longitudinal creases. Mucus-secreting cells present; in males, epithelial mucous glands form tubules extending into the lamina propria. Connective tissue of the lamina propria anchors the urethra to surrounding structures.
- **Clinical — Transitional cell carcinoma**: most common bladder cancer type; begins in the innermost transitional epithelial layer. Smoking is the most important risk factor (toxins filtered into urine, which sits in the bladder). Hallmark sign: **hematuria** (bloody urine). Diagnosed/treated via **cystoscopy** (thin lighted endoscope through the urethra into the bladder).

Urinary Reflexes: Urine Storage and Urine Voiding

- Urine reaches the bladder via ureteral peristalsis. Urge to urinate typically appears around **200 mL** of bladder volume. Continence vs. urination depends on interplay between spinal reflexes and higher (conscious) brain centers. Two spinal reflexes: **urine storage reflex** and **urine voiding reflex** — work together since storage and release involve coordinated, simultaneous activity.
- **Urine Storage Reflex**: mediated by spinal reflexes + the **pontine storage center**. Low-frequency afferent impulses from bladder stretch receptors: (1) increase sympathetic activity (inhibits detrusor contraction, stimulates internal urethral sphincter contraction); (2) stimulate contraction of the external urethral sphincter. The pontine storage center also stimulates somatic motor neurons to the external urethral sphincter, causing contraction. Net effect: promotes continence.
- **Urine Voiding Reflex**: mediated by spinal reflexes + the **pontine micturition center**. High-frequency afferent impulses from bladder stretch receptors stimulate interneurons relaying to the pontine micturition center, which initiates sacral spinal reflexes that: (1) increase parasympathetic activity (detrusor contracts, internal urethral

sphincter relaxes); (2) decrease sympathetic activity (internal urethral sphincter relaxes further); (3) decrease somatic motor activity (external urethral sphincter relaxes) → voiding occurs.

- At the end of a typical urination, <10 mL of urine remains in the bladder.
- **Clinical — Urinary tract infections (UTIs):** caused by bacteria, fungi, or viruses colonizing the urinary tract; **E. coli** most commonly implicated. Women are especially susceptible because of the short female urethra and its proximity to the anus; sexual intercourse can push bacteria toward the bladder. Symptoms: **dysuria** (painful urination), fever, foul/strong-smelling urine, cloudy or bloody urine, frequent urination, persistent urge to urinate; in the elderly, confusion may be the first sign. Diagnosed via bacteria/blood cells in urine. Inflammation of the urethra = **urethritis**; inflammation of the bladder lining = **cystitis** (both often co-occur). Treated with antibiotics; supportive care = dilute urine (drink fluids) and frequent voiding (empty bladder often). Untreated infection may ascend the ureters to the renal pelvis → **pyelitis** (inflammation of the renal pelvis); if bacteria invade the cortex/medulla too → **pyelonephritis** (high fever, intense flank pain, vomiting, diarrhea, blood cells/pus in urine).

26.3 | Aging and the Urinary System

- **Key Point:** aging produces four typical changes in the urinary system.
1. **Decline in functional nephrons:** total nephron count drops **30–40%** between ages 25 and 85.
 2. **Reduced glomerular filtration:** due to fewer glomeruli, cumulative filtration-apparatus damage in remaining glomeruli, and reduced renal blood flow.
 3. **Reduced sensitivity to ADH:** distal nephron and collecting system become less ADH-responsive → less water/sodium ion reabsorption → more frequent urination, increased daily fluid requirements.
 4. **Problems with urinary reflexes:** (a) external sphincter loses muscle tone, less effective voluntary retention → incontinence (often slow urine leakage); (b) loss of urination control from stroke, Alzheimer's disease, or other CNS problems affecting the cerebral cortex/hypothalamus; (c) in males, **urinary retention** may develop if an enlarged prostate compresses the urethra, restricting flow.

Ch 10 — Muscles of the Perineal Region and the Pelvic Diaphragm

10.5 | Muscles of the Perineal Region and the Pelvic Diaphragm

- **Key Point:** these muscles extend from the sacrum and coccyx to the ischium and pubis. They support the organs of the pelvic cavity, flex the joints of the sacrum and coccyx, and control the movement of materials through the urethra and anus.
- **Perineal region:** the pelvic floor and associated structures, bounded by the inferior margins of the pelvis.
- A line drawn between the **ischial tuberosities** divides the perineal region into two triangles:
 - **Urogenital triangle** (anterior).
 - **Anal triangle** (posterior).
- **Urogenital triangle layering:** superficial muscles = muscles of the external genitalia (bulbospongiosus, ischiocavernosus, superficial transverse perineal); these lie superficial to deeper muscles that strengthen the pelvic floor and encircle the urethra (deep transverse perineal muscle, external urethral sphincter).
- **Pelvic diaphragm:** an even more extensive muscular sheet that forms the muscular foundation of the **anal triangle** — composed of the **levator ani group** (**iliococcygeus** + **pubococcygeus**) and the **coccygeus** muscle.
- These pelvic floor muscles do **not** completely close the pelvic outlet: the **urethra, vagina, and anus** pass through them to open at the body surface. **Muscular sphincters** surround these openings and control voluntary urination and defecation. Muscles, nerves, and blood vessels also pass through the pelvic outlet en route to/from the lower limbs.

Muscle-by-muscle summary (Origin / Insertion / Action / Innervation), as given in the chapter table:

- **Bulbospongiosus (male):** Origin — median raphe / central tendon of the perineal region. Insertion — corpus spongiosum, perineal membrane, and corpus cavernosum. Action — compresses the base of and stiffens the penis; ejects urine or semen. Innervation — pudendal nerve, perineal branch (S2–S4).
- **Bulbospongiosus (female):** Origin — perineal body (central tendon of perineal region). Insertion — bulb of vestibule, perineal membrane, body of clitoris, and corpus cavernosum. Action — compresses and stiffens the clitoris; narrows the vaginal opening. Innervation — pudendal nerve, perineal branch (S2–S4).
- **Ischiocavernosus:** Origin — ramus and tuberosity of the ischium (also the ischiopubic ramus in females). Insertion — corpus cavernosum of the penis or clitoris. Action — compresses and stiffens the penis or clitoris, helping maintain erection. Innervation — pudendal nerve, perineal branch (S2–S4).
- **Superficial transverse perineal:** Origin — ischial ramus. Insertion — central tendon of the perineal region. Action — stabilizes the central tendon of the perineal region. Innervation — pudendal nerve, perineal branch (S2–S4).
- **Deep transverse perineal muscle:** Origin — ischial ramus. Insertion — perineal body. Action — stabilizes the central tendon of the perineal region. Innervation — pudendal nerve, perineal branch (S2–S4).
- **External urethral sphincter (male):** Origin — ischial and pubic rami. Insertion — median raphe at the base of the penis; inner fibers encircle the urethra. Action — closes the urethra; compresses the prostate and bulbo-urethral glands. Innervation — pudendal nerve, perineal branch (S2–S4).
- **External urethral sphincter (female):** Origin — ischial and pubic rami. Insertion — median raphe; inner fibers encircle the urethra. Action — closes the urethra; compresses the vagina and greater vestibular glands. Innervation — pudendal nerve, perineal branch (S2–S4).
- **Coccygeus** (pelvic diaphragm): Origin — ischial spine. Insertion — lateral, inferior borders (of the sacrum/coccyx). Action — flexes the coccygeal joints; elevates and supports the pelvic floor. Innervation — inferior sacral nerves.
- **Iliococcygeus** (levator ani group; pelvic diaphragm): Origin — ischial spine and pubis. Insertion — coccyx and median raphe. Action — tenses the floor of the pelvis, supports pelvic organs, flexes the coccygeal joints. Innervation — pudendal nerve (sacral segments).
- **Pubococcygeus** (levator ani group; pelvic diaphragm): Origin — inner margins of the pubis. Insertion — coccyx and median raphe. Action — tenses the floor of the pelvis, supports pelvic organs, flexes the coccygeal joints. Innervation — pudendal nerve (sacral segments).
- **External anal sphincter:** Origin — via tendon from the coccyx. Insertion — encircles the anal opening. Action — closes the anal opening. Innervation — pudendal nerve.
- Diagram labels noted in the figure: **Origin** = Pubococcygeus (part of the anal triangle floor); **Insertion** = urethral sphincter; landmark structures shown include the **anal triangle** and the **sacrotuberous ligament** (inferior male perineal view).

10.5 Concept Check

- Function of the perineal/pelvic diaphragm muscles: support pelvic organs, flex the sacrococcygeal joints, and control movement of materials through the urethra and anus (as stated in the Key Point above).

Ch 28 — Development and Inheritance (Placentation & Early Development)

Introduction

- **Development:** gradual modification of anatomical structures from conception to maturity — a single cell becomes a human body of trillions of cells organized into tissues, organs, organ systems.
- **Differentiation:** formation of specialized cell types during development, occurring through selective changes in genetic activity.

28.1 | An Overview of Development

- Development is a continuum divided into two periods:
 - **Prenatal development:** begins at fertilization/conception (fusion of sperm and oocyte), continues to delivery. **Embryology** = the study of developmental events during prenatal development.
 - **Postnatal development:** begins at birth, continues to maturity (full development/completed growth).
- Prenatal development is further subdivided:
 - **Pre-embryonic development:** first 2 weeks after fertilization; produces a pre-embryo that implants in the uterine lining.
 - **Embryonic development:** 3rd–8th weeks.
 - **Fetal development:** begins at the start of the 9th developmental week, continues until birth.
- Fertilization: fusion of two haploid gametes (23 chromosomes each) → **zygote** with 46 chromosomes (normal diploid number). Functional roles differ: sperm delivers paternal chromosomes only; the oocyte provides all nourishment and genetic programming to support embryonic development for nearly a week after conception — hence the oocyte's much greater volume (secondary oocyte diameter > 2x the entire length of a sperm at fertilization).

28.2 | Fertilization (content following 28.1, heading garbled in source but topic is clearly fertilization)

- Fertilization typically occurs near the **ampulla–isthmus junction** of the uterine tube, usually within a day of ovulation, during a 24-hour window (oocyte travels only a few cm in this time).
- Sperm arriving in the vagina are already motile but must undergo **capacitation** in the female tract before they can fertilize an oocyte.
- Sperm travel time from vagina to fertilization site: **30 minutes to 2 hours**, aided by uterine muscular contractions and ciliary currents in the uterine tubes.
- Of **~200 million sperm** in a typical ejaculate deposited in the vagina, only **~10,000** enter the uterine tube, and **fewer than 100** reach the isthmus. A sperm count below **20 million/mL** renders a male functionally sterile (too few surviving sperm to reach/fertilize the oocyte). Although only one sperm fertilizes the oocyte, dozens are required for successful fertilization overall.
- **Three complicating factors for fertilization:**
 1. The oocyte is surrounded by the **corona radiata** (layer of follicle cells).
 2. Metabolic operations of the oocyte have been discontinued.
 3. The oocyte leaving the follicle is arrested in **metaphase of meiosis II**.
 - The corona radiata protects the oocyte through the ruptured follicular wall into the infundibulum. A sperm must penetrate the corona radiata via the **acrosome reaction**, releasing acrosomal enzymes including **hyaluronidase**, which breaks down bonds between adjacent follicle cells. Dozens of sperm must release hyaluronidase before the corona radiata connections weaken enough for one intact sperm to reach the oocyte.
 - The successful sperm (with an intact acrosome) binds to **sperm receptors in the zona pellucida**, triggering acrosome rupture. On contact with the oocyte surface, the cell membranes fuse and the sperm enters the **ooplasm** (oocyte cytoplasm) — this is **oocyte activation**: metabolic rate rises suddenly, and immediate plasma membrane changes block fertilization by additional sperm (prevents **polyspermy**, which if it occurs prevents normal development).
 - Following activation, the oocyte completes meiosis; its nuclear material reorganizes as the **female pronucleus**. Simultaneously the sperm nucleus swells into the **male pronucleus**, migrates to the cell center, and the two pronuclei fuse — **amphimixis** — producing a zygote with the normal diploid complement of 46 chromosomes. This is the "moment of conception." The first **cleavage** division then yields two daughter cells, subdividing the zygote's cytoplasm.
- Fertilization steps sequence (from figure): (1) Oocyte at Ovulation — secondary oocyte + first polar body released, surrounded by corona radiata, arrested in metaphase II; (2) Fertilization and Oocyte Activation —

acrosomal enzymes from multiple sperm create gaps in the corona radiata, one sperm fuses with the oocyte membrane, triggering activation and completion of meiosis; (3) Pronucleus Formation Begins — sperm absorbed into cytoplasm, female pronucleus develops; (4) Spindle Formation and Cleavage Preparation — male pronucleus develops, spindle fibers appear; (5) Amphimixis Occurs and Cleavage Begins — first cleavage division nears completion ~30 hours after fertilization; (6) Cytokinesis Begins.

28.3 | Prenatal Development

- **Gestation:** the time spent in prenatal development; divided into **three trimesters**, each ~3 months:
 - **First trimester:** pre-embryonic, embryonic, and early fetal development; beginnings of all major organ systems appear.
 - **Second trimester:** development of fetal organs/organ systems; body proportions change; by its end the fetus looks distinctly human.
 - **Third trimester:** rapid fetal growth; early on, most major organ systems become fully functional; an infant born 1–2 months premature has a reasonable survival chance.

The First Trimester — Four general processes:

1. **Cleavage:** sequence of cell divisions beginning immediately after fertilization, ending when the zygote first contacts the uterine wall; the zygote becomes a pre-embryo that develops into a multicellular **blastocyst**.
2. **Implantation:** begins when the blastocyst attaches to the endometrium and continues as it burrows into the uterine wall; sets the stage for formation of vital embryonic structures.
3. **Placentation:** formation of blood vessels around the blastocyst edges — first step in forming the **placenta**, the structure linking maternal and embryonic systems, providing respiratory/nutritional support.
4. **Embryogenesis:** formation of a viable embryo; establishes the foundation for all major organ systems.
 - By the end of the first trimester (12th week), the fetus is ~**75 mm (3 in.)** long, weighs ~**14 g (0.5 oz)**. Only ~**40%** of conceptions produce embryos surviving the first trimester — hence advice to avoid drugs/alcohol/other disruptive stresses during this period.

Cleavage and Blastocyst Formation

- **Cleavage:** series of cell divisions subdividing the zygote's cytoplasm into progressively smaller daughter cells (**blastomeres**). First cleavage division completes ~**30 hours** after fertilization, producing a pre-embryo of 2 identical blastomeres. Subsequent divisions occur every **10–12 hours**; initially synchronous, later less predictable as blastomere number increases.
- After **3 days:** pre-embryo = solid ball of cells, the **morula** ("mulberry").
- After **5 days:** blastomeres form a hollow ball, the **blastocyst**, with an inner cavity, the **blastocoele**. Blastomeres are now unequal in size/shape.
- **Trophoblast:** outer cell layer separating the external environment from the blastocoele; provides food to the developing embryo; the only pre-embryo cells that contact the uterine wall.
- **Inner cell mass:** a second cell group clustered at one end of the blastocyst, exposed to the blastocoele on one side, insulated from the outside by the trophoblast on the other; these are stem cells responsible for producing all body cell types.

Implantation

- Immediately after fertilization, the zygote is **4 days** from the uterus; arrives in the uterine cavity as a morula; blastocyst formation occurs over the next **2–3 days**, during which cells absorb nutrients from glycogen-rich fluid secreted by endometrial glands within the uterine cavity fluid.
- When fully formed, the blastocyst contacts the endometrium and implantation occurs — usually in the **fundus or body of the uterus**.
- Implantation begins as the blastocyst surface closest to the inner cell mass touches/adheres to the uterine lining (**Day 7**). Trophoblast cells at the contact point divide rapidly (several layers thick). Near the endometrial wall, plasma membranes between trophoblast cells disappear, creating a multinucleate cytoplasmic layer — the **syncytiotrophoblast** (Day 8).

- The syncytiotrophoblast erodes through the uterine epithelium by secreting **hyaluronidase** (same enzyme mechanism as in sperm penetrating the corona radiata), dissolving proteoglycans between epithelial cells. This creates a gap that maternal epithelial cells later repair; once the blastocyst loses contact with the uterine cavity, further development occurs entirely within the endometrium's functional layer.
- As implantation proceeds (Days 8–9), the syncytiotrophoblast enlarges and spreads into the endometrium, disrupting and enzymatically digesting uterine glands; released nutrients are absorbed by the syncytiotrophoblast and diffuse across the underlying **cytotrophoblast** to the inner cell mass, fueling early embryo formation. Cytoplasmic extensions of the trophoblast grow around and destroy endometrial capillary walls; maternal blood flows slowly through trophoblastic channels called **lacunae**. Fingerlike **primary villi** (syncytiotrophoblast extension with a cytotrophoblast core) extend from the trophoblast into the endometrium. Over subsequent days the trophoblast breaks down larger endometrial veins/arteries, increasing lacunar blood flow.

Formation of the Blastodisc

- At implantation, the inner cell mass separates from the trophoblast, creating a fluid-filled **amniotic cavity**. By Days 10–11, the inner cell mass organizes into a two-layer oval sheet, the **blastodisc**: the **epiblast** (facing the amniotic cavity) and the **hypoblast** (exposed to the blastocoele contents).

Gastrulation and Germ Layer Formation

- **Gastrulation** (Day 12): a third layer forms via cell migration. Epiblast cells migrate toward the blastodisc's center, toward the **primitive streak**, then move between the epiblast and hypoblast, creating three distinct **germ layers**. After gastrulation begins: the layer remaining in contact with the amniotic cavity = **ectoderm**; the hypoblast becomes the **endoderm**; the new middle layer = **mesoderm**.
- Gastrulation produces the two-layered **embryonic disc**, which forms the embryo's body; all other blastocyst cells become extra-embryonic membranes.
- **Germ layer contributions** (organogenesis foundations):
 - **Ectoderm**: integumentary system (epidermis, hair follicles/hairs, nails, sweat/sebaceous/mammary glands communicating structures); skeletal system (pharyngeal cartilage derivatives — part of sphenoid, auditory ossicles, styloid processes of temporal bones, hyoid horns/superior rim); all nervous tissue (brain, spinal cord); endocrine (pituitary gland, adrenal medullae); respiratory (mucous epithelium of nasal passageways); digestive (mucous epithelium of oral cavity and anus, salivary glands). *Neural crest (ectodermal) contributes to skull formation and skeletal derivatives of embryonic pharyngeal arches.*
 - **Mesoderm**: integumentary (dermis, except epidermal derivatives); skeletal system (all components except some pharyngeal derivatives); muscular system (all components); endocrine (adrenal cortex, endocrine tissue of heart/kidneys/gonads); cardiovascular system (all components incl. bone marrow); lymphatic system (all components); urinary system (kidneys incl. nephrons and initial collecting system portions); reproductive system (gonads and adjacent duct system portions); miscellaneous (linings of thoracic/pleural/pericardial/peritoneal cavities, connective tissues supporting all organ systems).
 - **Endoderm**: endocrine (thymus, thyroid, pancreas); respiratory (respiratory epithelium except nasal passageways, and associated mucous glands); digestive (mucous epithelium except mouth/anus, exocrine glands except salivary glands, liver, pancreas); urinary (urinary bladder, distal duct system portions); reproductive (distal duct system portions; stem cells producing gametes).

Formation of Extra-embryonic Membranes — four structures supporting embryonic/fetal development by maintaining a stable environment and providing access to maternal oxygen/nutrients; leave few traces in adult systems:

1. **Yolk sac** (endoderm + mesoderm): first extra-embryonic membrane to develop; migrating hypoblast cells form a pouch below the blastodisc, visible ~10 days after fertilization; mesodermal cells complete its formation during gastrulation; blood vessels appear within the mesoderm, making the yolk sac an important early site of **blood cell formation**.
2. **Amnion** (ectoderm + mesoderm): ectodermal cells spread over the inner amniotic cavity surface, followed by mesodermal cells forming an outer layer. Expands as the embryo/fetus enlarges, increasing amniotic cavity size. Contains **amniotic fluid**, which surrounds and cushions the developing embryo/fetus.

3. **Allantois** (endoderm + mesoderm): begins (Week 3) as an outpocketing of endoderm near the yolk sac base; the free endodermal tip grows toward the blastocyst wall, surrounded by mesodermal cells; its base later gives rise to the **urinary bladder**.

4. **Chorion** (mesoderm + trophoblast): mesoderm associated with the allantois spreads around the entire inside of the trophoblast, forming a mesodermal layer beneath it — combination of mesoderm + trophoblast = chorion. Solves the diffusion-distance problem as the embryo enlarges: blood vessels developing within the chorionic mesoderm link the embryo with the trophoblast; circulation through chorionic vessels begins early in the third week, when the heart starts beating.

- As the embryo and extra-embryonic membranes bulge into the uterine cavity, the trophoblast (still endometrium-covered) no longer participates in nutrient absorption/embryo support. The embryo moves away from the placenta; the **body stalk** (connects embryo to chorion; contains distal allantois and placental blood vessels) and **yolk stalk** (narrow endoderm-to-yolk-sac connection) fuse to form the **umbilical stalk**.

Placentation

- **Placentation**: formation of the placenta from embryonic and maternal tissues; the appearance of blood vessels in the chorion is the first step toward a functional placenta.
- By the third week, mesoderm extends along the core of each trophoblastic villus, forming **chorionic villi** in contact with maternal tissue; these enlarge/branch into an intricate endometrial network. Maternal blood vessels continue to be eroded; maternal blood flows slowly through lacunae lined by syncytiotrophoblast. **Diffusion** occurs between maternal blood in the lacunae and fetal blood in vessels within the chorionic villi.
- Initially the entire blastocyst is surrounded by chorionic villi; the chorion continues enlarging like a balloon within the endometrium.
- **Regional endometrial differentiation** as the placenta expands:
 - **Capsular decidua (decidua capsularis)**: thin endometrial portion covering the embryo, separating it from the uterine cavity; no longer exchanges nutrients; chorionic villi disappear here.
 - **Basal decidua (decidua basalis)**: disc-shaped region in the deepest endometrium where placental functions concentrate.
 - **Parietal decidua (decidua parietalis)**: the rest of the endometrium, with no chorionic contact.
- By week 4, the embryo/amnion/yolk sac are suspended within an expansive fluid-filled chamber.
- **Placental circulation** (near end of first trimester): blood flows fetus → placenta via paired **umbilical arteries**, returns to the fetus via a single **umbilical vein**. Active/passive exchange between fetal and maternal bloodstreams occurs at the **chorionic villi**.
- **Placental hormones**: **Human chorionic gonadotropin (HCG)** production begins a few days after implantation; stimulates the corpus luteum to keep producing progesterone through early pregnancy. During the second/third trimesters, the placenta also releases **progesterone**, **estrogens**, **human placental lactogen (HPL)**, and **relaxin** — all synthesized/released into maternal circulation by the trophoblast.
- Uterine growth over gestation: from 7.5 cm (3 in.) to **30 cm (12 in.)** long, containing almost **5 L** of fluid at term; uterus + contents weigh roughly **10 kg (22 lb)**. This expansion occurs via enlargement/elongation of existing smooth muscle fibers (not new fiber formation) over 16 weeks to full term (9 months). At term, the enlarged uterus/fetus displaces many abdominal organs from their normal positions.

Embryogenesis

- Shortly after gastrulation begins, the embryo's body separates from the rest of the embryonic disc via folding and differential growth, producing a bulge projecting into the amniotic cavity — the **head fold**. Similar movements form a **tail fold** (Weeks 3–4). The embryo becomes physically/developmentally separated from the rest of the blastodisc and extra-embryonic membranes; anterior/posterior surfaces and left/right sides become visible.
- The first trimester is critical because these events establish the basis for organ formation — **organogenesis**.

The Second and Third Trimesters

- **Key Point**: fetus grows rapidly; all major organ systems complete development and become functional.

- By end of first trimester, foundations of all major organ systems have formed; during the second trimester these systems complete functional development. By end of second trimester, fetus weighs ~**0.64 kg (1.4 lb)**; the fetus (encircled by amnion) grows faster than the surrounding placenta; the amnion's mesodermal outer covering fuses with the chorion's inner lining.
- During the third trimester, all fetal organ systems become functional; growth rate slows somewhat, but this trimester sees the **greatest weight gain** — fetus gains ~**2.6 kg (5.7 lb)** in the last 3 months, reaching full-term weight of ~**3.2 kg (7.05 lb)**.

Organogenesis milestones by gestational month (selected highlights from the chapter's summary table; "(b)" = begins formation, "(c)" = completes formation):

- **Month 1** (5 mm, 0.02 g): somite formation (skeletal/muscular); neural tube formation begins; eye and ear formation begins; heartbeat begins; trachea/lung formation begins; intestinal formation begins; allantois formation begins.
- **Month 2** (1.6 cm, 1.0 g): nail beds/hair follicles/sweat glands begin forming; axial/appendicular musculature formation begins; CNS/PNS organization, cerebrum growth begins; taste buds/olfactory epithelium begin; thymus/thyroid/lymph node formation begins; basic heart structure and major blood vessels complete; bronchial subdivisions/ducts extend; kidney formation begins; mammary gland (adult form) formation begins.
- **Month 3** (5.4 cm, 14 g): epidermal layers appear; ossification centers spread; rudiments of appendicular musculature complete; basic spinal cord/brain structure complete; thymus/thyroid gland complete; tonsils, blood formation in red bone marrow begins; gallbladder/pancreas complete; definitive gonads/ducts/genitalia formation begins.
- **Month 4** (11.6 cm, 100 g): hair/sebaceous glands begin forming, sweat glands complete; articulations begin, facial/palatal organization completes; fetus starts moving; rapid cerebral expansion; basic eye/ear structure completes, peripheral receptor formation begins; lymphocyte migration to lymphoid organs begins; embryonic kidneys degenerate.
- **Month 5** (16.4 cm, 300 g): keratin/nail production begins; spinal cord myelination begins; tonsils complete; nostrils open; intestinal subdivisions complete.
- **Month 6** (30 cm, 600 g): perineal muscles complete; CNS tract formation begins; adrenal glands complete; spleen/liver/bone marrow blood formation complete; alveolar formation begins; epithelial organization completes.
- **Month 7** (37.6 cm, 1.005 kg): keratinization, nail formation begins; eyelids open, retina complete; pituitary gland complete; intestinal plicae complete; descent of testes begins.
- **Month 8** (42.4 cm, 1.702 kg): epiphyseal cartilage formation completes; taste receptors complete; pulmonary branching complete and functional; complete nephron formation; descent of testes completes at or near birth.
- **Postnatal:** hair changes in consistency/distribution continue; epiphyseal cartilage formation/growth continues; muscle mass/control increase; myelination/layering/CNS tract formation continue.
- (Full Embryology Summaries for each organ system are cross-referenced at the end of the chapter, e.g., Digestive System pp. 781–782, Urinary System pp. 783–784, Reproductive System pp. 785–787, Heart p. 775, Respiratory System pp. 779–780.)
- **Clinical — Congenital malformations:** structural/functional anomalies present at birth. Worldwide, 94% of severe congenital malformations occur in low/middle-income countries. Up to half of all conceptions produce zygotes that don't survive cleavage; at least 10% of fertilizations produce zygotes with abnormal chromosomes (most never complete development). Of embryos that implant, ~20% fail to complete 5 months of development (average survival ~8 weeks) — this pregnancy failure = **spontaneous abortion (miscarriage)**. Less than 1% of newborns show chromosomal abnormalities from spontaneous mutations. Common malformations: congenital heart disease, Down syndrome, cleft lip/palate, spina bifida, clubfoot deformity, **phenylketonuria (PKU)** (absence/deficiency of the enzyme processing phenylalanine). Of 2.75 million annual worldwide neonatal deaths, 10% are due to congenital malformations. **Consanguinity** (biological parents related by blood) increases rare genetic condition prevalence and nearly doubles neonatal death risk. Maternal infections (rubella, syphilis, Zika) can cause malformations. Maternal nutritional status (folate/iodine deficiency, obesity, diabetes mellitus) linked to congenital problems — folate deficiency increases neural tube defect risk (spina bifida, anencephaly). Environmental risk factors: maternal alcohol/tobacco/certain prescription meds/street drugs/radiation exposure. Advanced maternal and paternal age increase risk of chromosomal conditions like

Down syndrome (**trisomy 21**, extra chromosome 21). Screening options: preconception screening; **amniocentesis** (testing amniotic fluid for fetal cell abnormalities); **chorionic villus sampling** (testing chorionic villi for fetal cells) — both performable in 1st/2nd trimester; neonatal (newborn) screening for blood disorders, metabolic/hormone defects, congenital deafness, heart defects. Public health approach requires surveillance, causal research, and prevention research/programs.

- **Clinical — Teratogens:** any agent causing embryo/fetal abnormality during prenatal development; fetus most vulnerable during the **first trimester**. Categories: physical agents (high fever), maternal health factors (diabetes, obesity, hypertension), environmental chemicals (herbicides, industrial solvents), infectious agents (rubella, Zika). In the U.S., the most frequent cause of *preventable* abnormal fetal development is maternal tobacco/alcohol/drug use. Smoking nearly doubles risk of low-birth-weight delivery (poor prenatal growth and/or premature delivery). **Fetal alcohol syndrome (FAS):** most common known nongenetic cause of growth deficiency, functional deficits, intellectual impairment; alcohol crosses the placenta and is metabolized more slowly in the fetus, so fetal alcohol levels stay elevated longer. FAS features: small heads, characteristic facial abnormalities, short stature, low weight, poor fine/gross motor coordination, speech/language delays, attention/memory deficits, vision/hearing problems, heart/kidney/bone abnormalities — effects vary by child but are irreversible. Maternal use of methamphetamine, heroin, codeine, oxycodone, methadone crosses the placenta and can cause fetal addiction; after birth, newborns develop **neonatal abstinence syndrome** (withdrawal): hyperreflexia, increased muscle tone, irritability, poor feeding, vomiting, diarrhea, sleep difficulty, seizures; associated CNS abnormalities (learning/behavioral difficulties) can last a lifetime.
- **Clinical — Multiple births:** increasing due to fertility drugs/assisted reproductive technology and rising average maternal age (older women more likely to release >1 oocyte per cycle). US rate: 33.7 twin births per 1,000 deliveries. **Dizygotic ("fraternal") twins:** develop from two separate oocytes ovulated and fertilized; odds against two zygotes from the same parents having identical genes exceed 1 in 8.4 million. **Monozygotic ("identical") twins:** result from separation of blastomeres early in cleavage, or splitting of the inner cell mass before gastrulation — genetically identical since both formed from the same gamete pair. Triplets/quadruplets/larger multiples arise from multiple ovulations, blastomere splitting, or a combination. Natural multiple-birth rates: twins 1 in 89 births, triplets 1 in 7,921, quadruplets 1 in 704,969. Fertility drugs stimulating maturation of abnormally high follicle numbers increase multiple-birth incidence.

28.4 | Labor and Delivery (Parturition) — heading garbled in source but content is clearly this topic

- **Parturition (childbirth):** forcible expulsion of the fetus from the uterus through the cervix and vagina, via a series of strong rhythmic uterine contractions called **labor**. Labor is triggered by increased oxytocin levels combined with increased uterine sensitivity to oxytocin, stimulating myometrial contractions. **True labor** contractions begin near the uterine fundus and sweep in a wave toward the cervix, are strong, and occur at regular intervals (contrast with occasional, irregular **false labor** spasms).

Stages of Labor — Key Point: divided into three stages — dilation, expulsion, and placental.

1. **Dilation stage:** begins with true labor onset; cervix dilates, fetus moves down the cervical canal; highly variable length, typically **8+ hours**. Contractions initially last up to 30 seconds, occurring every 10–30 minutes. Late in this stage the amnion usually ruptures ("water breaking").

2. **Expulsion stage:** begins as the cervix, pushed open by the approaching fetus, completes dilation to about **10 cm (4 in.)**; continues until the fetus fully emerges from the vagina — usually **<2 hours**. Delivery/birth = the newborn's arrival into the outside world. If the vaginal canal is too small (risking perineal tearing), an **episiotomy** (incision through the perineal musculature) may be performed to temporarily enlarge the passage — repaired with sutures afterward (simpler than repairing a jagged tear). If progress is slow or complications arise, delivery may proceed by **cesarean section (C-section)** — incision through the abdominal wall and uterus to remove the infant.

3. **Placental stage:** muscle tension builds in the walls of the now partially empty (and shrinking) uterus; this contraction tears the endometrium–placenta connections. Usually within an hour after delivery, the placenta (afterbirth) is ejected. Placental disruption causes blood loss of **~500–600 mL**, easily tolerated given the pregnancy-related increase in maternal blood volume.

Premature Labor — Key Point: occurs when true labor begins before the fetus has completed normal development.

- Newborn survival chances correlate directly with birth weight. Most fetuses born at **25–27 weeks** gestation (birth weight <600 g/21.1 oz) die despite intensive neonatal care; survivors have high risk of developmental abnormalities. **Premature delivery** usually refers to birth at **28–36 weeks** (birth weight >1 kg/2.2 lb) — with care, good chance of normal survival/development.

Clinical — Common Complications of Parturition:

- **Fetal distress:** inadequate fetal oxygenation, typically shown by decreased fetal heart rate; may require emergency C-section or assisted delivery (forceps/vacuum extractor).
- **Breech position (malpresentation):** baby positioned buttocks/feet first; risk of getting "stuck" since the head (largest part) comes last — indication for C-section.
- **Placenta previa:** placenta covers the cervix; if labor proceeds naturally the placenta can separate too early, causing potentially deadly maternal/fetal blood loss — often managed with a scheduled C-section before natural labor begins.
- **Meconium aspiration:** fetal intra-uterine aspiration of amniotic fluid contaminated with **meconium** (first feces), causing breathing difficulty. Risk factors: post-date ("aging") placenta, decreased fetal oxygenation in utero, maternal diabetes or hypertension, long/difficult labor.
- **Cephalopelvic disproportion (CPD):** fetal head too large to pass through the maternal pelvis, preventing vaginal delivery — the most common reason for a C-section.

28.5 | The Neonatal Period

- **Key Point:** during the neonatal period, the neonate continues undergoing physiological and anatomical changes.
- **Neonatal period:** from birth to 1 month after. The newborn has few of the anatomical/functional/physiological characteristics of a mature adult; developmental processes continue after delivery.
- Before delivery, dissolved gases, nutrients, wastes, hormones, and immunoglobulins were exchanged across the placental interface; at birth the newborn must become relatively self-sufficient. Key transitions from fetus to neonate:

1. **Respiratory:** at birth, lungs are collapsed and fluid-filled; filling them with air requires a massive, powerful first inhalation.
2. **Cardiovascular:** lung expansion alters blood pressure/flow patterns; the **ductus arteriosus** closes, isolating the pulmonary and systemic trunks; the **foramen ovale** closes, separating the atria — completing separation of the pulmonary and systemic circuits.
3. **Vital signs:** neonatal heart rate **120–140 bpm** and respiratory rate **30 breaths/min** — considerably higher than adult rates; both are normal for neonates.
4. **Digestive:** relatively inactive before birth, though it accumulates **meconium** (mixture of bile secretions, mucus, epithelial cells), excreted in the first few days of life, during which the newborn begins to nurse.
5. **Renal/fluid balance:** as arterial waste builds up, the kidneys excrete it; glomerular filtration is normal, but neonatal kidneys cannot significantly concentrate urine — resulting in high urinary water losses and proportionally much greater neonatal fluid requirements than adults.
6. **Thermoregulation:** in the first few days, the neonate has little ability to control body temperature; as the infant grows and its subcutaneous adipose "insulating blanket" thickens, metabolic rate rises; daily/hourly body temperature fluctuations continue throughout childhood.

Embryology Summary (Integumentary System Development) — appears at end of extracted range (p.756)

- Skin development proceeds month-by-month: Month 3 — epidermal basal cell divisions form cords of cells growing into the dermis (become hair follicles, sebaceous glands, sweat glands), surrounded by mesenchymal cells; embryonic connective tissue differentiates into the dermis (fibroblasts, increasing collagen fiber density; deeper region becomes dense irregular collagen network) and, below it, the subcutaneous layer (loose connective tissue).

- Month 4 — nails begin as epidermal thickenings near fingertip/toe tips, settling into the dermis with a distinct border; nail production initially involves all basal cells of the nail field, later (by birth) restricted to the nail root.
- As basal cell divisions continue, the epidermal layer thickens, forms irregular folds; **melanocytes** (pigment cells) migrate in and squeeze between basal cells — the epithelium comes to resemble adult epidermis.
- Month 5 — hair follicle develops as a deep epithelial column surrounding a **papilla** (small connective tissue mass); hair growth occurs in epithelium covering the papilla; an epithelial outgrowth forms a sebaceous gland; a sweat gland develops as an epithelial column elongates, coils, and becomes hollow. Mammary glands develop comparably but with much broader, more extensive branching.
- At birth: a hair projects from the follicle, sebaceous gland secretions lubricate the hair shaft; sweat gland ducts carry secretions to the skin surface; mammary glands have not completed development — in females, further ductal/glandular elaboration occurs at puberty, with functional maturity not reached until late pregnancy.

Ch 17 — The Autonomic Nervous System (Hypogastric/Pelvic Autonomic Plexuses)

17.3/17.4 | Receptor types and neurotransmission (context preceding the plexus discussion)

- **Parasympathetic neuroeffector junctions** are small, with narrow synaptic clefts; effects are short-lived because most released acetylcholine (ACh) is inactivated by **acetylcholinesterase (AChE)** within the synapse (and any ACh diffusing into surrounding tissue is likewise deactivated by AChE) — so parasympathetic stimulation effects are localized and brief.
- **Nicotinic receptors:** located on the surfaces of all ganglionic neurons of both parasympathetic and sympathetic divisions, and at somatic neuromuscular synapses. ACh exposure always causes **excitation** via opening of chemically gated Na⁺ channels in the postsynaptic membrane.
- **Muscarinic receptors:** found at all cholinergic neuromuscular/neuroglandular junctions in the parasympathetic division, and at the few cholinergic neuroeffector junctions in the sympathetic division. Produce longer-lasting effects than nicotinic stimulation; response (excitatory or inhibitory) reflects activation/inactivation of specific enzymes.
- Names reflect the stimulating compounds: nicotinic receptors bind nicotine; muscarinic receptors bind muscarine.

Summary of the Parasympathetic Division

- Includes visceral motor nuclei in the brainstem associated with **four cranial nerves (III, VII, IX, X)**. Autonomic nuclei also lie in the lateral portions of the anterior horns in **sacral segments S2–S4**.
- Ganglionic neurons are located in **terminal or intramural ganglia**.
- Innervates structures in the head and organs in the thoracic and abdominopelvic cavities.
- All parasympathetic neurons are **cholinergic**. Preganglionic ACh release stimulates nicotinic receptors on ganglionic neurons — always excitatory. ACh release at neuroeffector junctions stimulates muscarinic receptors — effect (excitatory or inhibitory) depends on enzymes activated upon receptor binding.
- Effects of parasympathetic stimulation are brief and restricted to specific organs/sites.

17.4 | Relationship between the Sympathetic and Parasympathetic Divisions

- **Sympathetic division:** widespread impact, reaching visceral organs/tissues throughout the body.
- **Parasympathetic division:** modifies activity of structures innervated by specific cranial nerves and pelvic nerves, including thoracic and abdominopelvic visceral organs.
- Some organs receive innervation from only one division; most vital organs receive **dual innervation** (both sympathetic and parasympathetic).

- Where dual innervation exists, the two divisions often have **antagonistic (opposite)** effects. Dual innervation is most common in the **digestive tract, heart, and lungs**. Example: sympathetic stimulation **decreases** digestive tract motility; parasympathetic stimulation **increases** it.

Anatomy of Dual Innervation

- **In the head:** parasympathetic postganglionic fibers from the **ciliary, pterygopalatine, submandibular, and otic ganglia** accompany cranial nerves to their peripheral destinations; sympathetic innervation reaches the same structures by traveling directly from the **superior cervical ganglia** of the sympathetic chain.
- **In the thoracic and abdominopelvic cavities:** sympathetic postganglionic fibers intermix with parasympathetic preganglionic fibers at a series of **plexuses**: the **cardiac plexus, pulmonary plexus, esophageal plexus, celiac plexus, inferior mesenteric plexus, and hypogastric plexus**. Nerves leaving these plexuses travel with the blood vessels and lymphatics supplying visceral organs.
- **Thoracic cavity:** autonomic fibers intersect at the **cardiac plexus** and **pulmonary plexus** — both contain sympathetic fibers innervating the heart and parasympathetic fibers innervating the heart and lungs. The **esophageal plexus** contains descending vagus nerve branches and splanchnic nerves leaving the sympathetic chain ganglia on each side.
- **Abdominopelvic cavity:** parasympathetic preganglionic fibers of the **vagus nerve** follow the esophagus into the abdominopelvic cavity and join the **celiac plexus (solar plexus)** network. The celiac plexus and the smaller associated **inferior mesenteric plexus** innervate abdominal cavity viscera.
- **Hypogastric plexus:** contains (1) the parasympathetic outflow of the **pelvic nerves**, (2) sympathetic postganglionic fibers from the **inferior mesenteric ganglion**, and (3) **sacral splanchnic nerves** from the sympathetic chain. **The hypogastric plexus innervates the digestive, urinary, and reproductive organs of the pelvic cavity** — i.e., this is the key autonomic plexus for the pelvis/perineum.

Visceral Reflexes

- **Visceral reflexes:** autonomic reflexes initiated in the viscera; provide automatic motor responses that can be modified, facilitated, or inhibited by higher centers, especially the **hypothalamus**. Example: pupillary constriction/dilation nuclei are also controlled by hypothalamic centers tied to emotional states — pupils constrict when queasy/nauseated, dilate when sexually aroused.
- Visceral reflex arcs have the same basic components as somatic reflexes (receptor, sensory nerve, etc.) but **all visceral reflexes are polysynaptic**.
- **Visceral sensory neurons** deliver info to the CNS via dorsal roots of spinal nerves, sensory branches of cranial nerves, and autonomic nerves innervating visceral effectors.
- **Long reflexes:** resemble polysynaptic somatic reflexes; processing involves CNS interneurons; motor neurons located in the brainstem or spinal cord; the ANS carries motor commands to visceral effectors after a synapse in a peripheral autonomic ganglion.
- **Short reflexes:** bypass the CNS entirely; involve sensory neurons and interneurons with somas located within autonomic ganglia; interneurons synapse on ganglionic neurons, motor commands distributed via postganglionic fibers; control very simple, localized motor responses.
- In general: short reflexes control activity patterns in one part of a target organ; long reflexes coordinate activity of the entire organ.

Sympathetic vs. Parasympathetic Division comparison (summary table):

Characteristic	Sympathetic	Parasympathetic
Location of CNS visceral motor neurons	Lateral horns of spinal segments T1–L2	Brainstem and spinal segments S2–S4
Location of PNS ganglia	Sympathetic chain ganglia; collateral ganglia (celiac, superior mesenteric, inferior mesenteric) anterior/lateral to the descending aorta	Within target organ tissues (intramural ganglia) or close to the target organ (terminal ganglia)
Preganglionic fiber length	Relatively short, myelinated	Relatively long, myelinated

Postganglionic fiber length	Relatively long, unmyelinated	Relatively short, unmyelinated
Neurotransmitter released	Usually norepinephrine	Always acetylcholine
Neuroeffector junction	Varicosities and enlarged terminal knobs releasing transmitter near target cells	Junctions releasing transmitter to specialized receptor surfaces
Divergence (CNS to ganglion cells)	~1:32	~1:6
General functions	Stimulate metabolism, increase alertness, prepare for emergency ("fight or flight")	Promote relaxation, nutrient uptake, energy storage ("rest and repose/digest")

Selected visceral reflexes relevant to the pelvis (from the chapter's reflex summary table):

- **Defecation reflex:** stimulus = distention of the rectum; response = relaxation of the internal anal sphincter; requires voluntary relaxation of the external anal sphincter to complete.
- **Urination reflex:** stimulus = distention of the urinary bladder; response = contraction of urinary bladder walls and relaxation of the internal urethral sphincter; requires voluntary relaxation of the external urethral sphincter.
- **Sexual arousal:** stimulus = erotic stimuli (visual or tactile); response = increased glandular secretions, increased sensitivity.
- **Emission and ejaculation (males, sympathetic reflex):** stimulus = erotic (tactile) stimuli; response = contraction of the seminal glands and prostate plus skeletal muscle contractions that eject semen; ejaculation itself involves contractions of the **bulbospongiosus muscles**.
- (Also listed among general visceral/sympathetic reflexes for context: gastric/intestinal reflexes, swallowing, vomiting, coughing, baroreceptor reflex, cardioacceleratory reflex, vasomotor reflexes, pupillary reflex, direct/consensual light reflexes — mediated by cranial nerves III and X and various medullary centers.)

Clinical Note — Urinary Bladder Dysfunction Following Spinal Cord Injury

- Normal urination requires coordinated visceral (autonomic) reflexive **detrusor contraction** and **relaxation of the involuntary internal urethral sphincter**, accompanied by voluntary (somatic) relaxation of the external urethral sphincter.
- Afferent sensory nerves from bladder-wall stretch receptors enter the spinal cord between **L1 and S4**. Efferent **parasympathetic nerves S2–S4** innervate the detrusor and the internal (involuntary) sphincter — increased parasympathetic activity causes bladder contraction and internal sphincter relaxation, producing the urge to urinate. Efferent **sympathetic nerves L1–L2** inhibit bladder contraction and cause the involuntary urethral sphincter to contract, preventing urine passage. The **puddental nerves (S2–S4)** of the somatic nervous system regulate voluntary relaxation of the external urethral sphincter's skeletal muscle, allowing urination.
- After a spinal cord injury (SCI), the kidneys are unaffected, but communication between bladder and brain is disrupted. During **spinal shock** (lasting days to several weeks), there is no somatic or visceral bladder activity — the bladder is completely **flaccid**, requiring continuous catheter drainage.
- After spinal shock subsides, bladder/sphincter function depends on injury level/completeness:
 - Injury at **T12 or above**: results in a **reflexive, spastic bladder** that empties without voluntary control.
 - Disruption of the reflex arc in spinal nerves **S2–S4**: results in a **flaccid bladder** that cannot contract and therefore cannot empty.
- Involuntary urination = **incontinence**. Urinary tract health after SCI requires an indwelling urinary catheter or intermittent catheterization to empty the bladder.

Ch 22 — Blood Vessels and Circulation (Internal/External Iliac Vessels)

The Descending Aorta / Abdominal Aorta (context leading to the iliac vessels)

- The **descending aorta** is continuous with the aortic arch; the diaphragm divides it into the superior **thoracic aorta** and inferior **abdominal aorta**.

- **Thoracic aorta:** begins at vertebra T1 level, lies slightly left of midline, gradually shifts to midline, exits through the posterior mediastinum, passes through the diaphragm at vertebra **T12**. Supplies thoracic organs (except the heart), chest/diaphragm muscles, and the thoracic spinal cord. Branches: visceral (bronchial, pericardial, mediastinal, esophageal arteries) and parietal (intercostal arteries — chest muscles/vertebral column area).
- **Abdominal aorta:** begins immediately inferior to the diaphragm, descends slightly left of the vertebral column, posterior to the peritoneal cavity, often cushioned by adipose tissue. At vertebra **L4** it splits into the right and left **common iliac arteries** — this split point is the **terminal segment of the aorta**.
- Abdominal aorta delivers blood to all abdominopelvic organs/structures. Unpaired visceral branches (anterior surface, into mesenteries): **celiac trunk, superior mesenteric artery, inferior mesenteric artery**. Paired branches (lateral surface, to body wall/kidneys/etc.): **inferior phrenic, adrenal, renal, gonadal, lumbar arteries**.
 - **Gonadal arteries:** originate between the superior and inferior mesenteric arteries. Males: **testicular arteries** — long, thin, supply the testes and scrotum. Females: **ovarian arteries** — supply the ovaries, uterine tubes, and uterus. (Distribution of gonadal vessels, both arteries and veins, differs by sex.)
- **Clinical — Aortic aneurysm:** a bulge/weakness in an arterial wall; can occur anywhere including the aorta. Classified by location: thoracic (ascending or descending) or abdominal. Often asymptomatic early, discovered incidentally on imaging (though sometimes palpable/auscultable in thin patients). Screening recommendation (per some primary care physicians): one-time ultrasound for people >65 with a family history, and men >65 who have ever smoked. If ruptured, prognosis is very poor — mortality rate >**90%**. Preventive surgery sometimes recommended for aneurysms >**5 cm** diameter that are enlarging. Open repair: midline incision, aorta cross-clamped above/below the aneurysm, synthetic tube graft sutured across the defect. Less-invasive **endovascular repair:** small femoral artery incision, guide wire/sheath advanced to the aneurysm site, stent graft inserted via the femoral artery and expanded/attached to the aortic wall to prevent further expansion.

Arteries of the Pelvis and Lower Limbs (Internal/External Iliac Arteries)

- At vertebra **L4**, the abdominal aorta's terminal segment divides into the **right and left common iliac arteries** plus the small **median sacral artery**. These carry blood to the pelvis and lower limbs.
- **Common iliac arteries:** travel along the inner surface of the ilium, descending posterior to the cecum and sigmoid colon. At the **lumbosacral joint**, each common iliac divides into an **internal iliac artery** and an **external iliac artery**.
- **Internal iliac artery:** enters the pelvic cavity; supplies the **urinary bladder**, internal/external walls of the pelvis, external genitalia, and the medial side of the thigh. Major branches: **superior gluteal, internal pudendal, obturator, and lateral sacral arteries**. In females, these vessels also supply the **uterus and vagina**.
- **External iliac artery:** supplies blood to the lower limbs; much larger in diameter than the internal iliac artery.
- **Arteries of the thigh/leg (continuation of the external iliac):** the external iliac artery crosses the iliopsoas, penetrates the abdominal wall, exits between the anterior superior iliac spine and pubic symphysis, and emerges on the anteromedial thigh as the **femoral artery**. ~5 cm distal to its exit, the **deep femoral artery** branches off laterally; **medial and lateral circumflex arteries** branch from the deep femoral artery — together the femoral, deep femoral, and circumflex arteries supply the thigh muscles and ventral/lateral thigh skin. The femoral artery continues posterior to the femur; at the popliteal fossa it gives off the **descending genicular artery** (supplies the medial knee), then becomes the **popliteal artery** as it passes through the adductor magnus. The popliteal artery crosses the popliteal fossa and branches into the **posterior tibial artery** and **anterior tibial artery**. The posterior tibial artery gives rise to the **fibular (peroneal) artery** ~4 cm distal to its formation.
- **Arteries of the foot:** the anterior tibial artery becomes the **dorsalis pedis artery** at the ankle, which branches to supply the ankle/dorsal foot. The posterior tibial artery divides at the ankle into the **medial and lateral plantar arteries** (supply the plantar foot surface); these connect to the dorsalis pedis via anastomoses linking the **dorsal (arcuate) arch** to the **plantar arch**.

Veins of the Lower Limb, Pelvis, and Abdomen (Internal/External Iliac Veins)

- General venous principles: veins collect blood from tissues/organs, draining ultimately into the superior or inferior vena cava (except cardiac veins → coronary sinus). Arteries and veins often run together with matching

names and share a connective tissue sheath with same-named nerves. Two key arterial-venous differences in the neck/limbs: (1) venous drainage is variable (figures show the most common pattern); (2) neck/limb arteries are deep (protected by bone/soft tissue), while neck/limbs typically have **two sets of peripheral veins — one superficial, one deep** — important for thermoregulation (cold: superficial veins bypassed, blood flows via deep veins to conserve heat; heat: skin blood supply increases, superficial veins dilate — why superficial arm/leg veins become prominent during heavy exercise).

- **Veins draining the lower limb:** plantar veins (sole) → deep leg veins (anterior tibial, posterior tibial, fibular/peroneal veins). Dorsal venous arch (dorsal foot) drained by two superficial veins: the **great saphenous vein** (longest vein in the body; travels proximally along the medial leg/thigh, drains into the femoral vein near the hip; commonly harvested for coronary artery bypass grafts) and the **small saphenous vein** (arises from the dorsal venous arch, travels along the posterior/lateral calf, enters the popliteal fossa, drains into the popliteal vein). The **popliteal vein** forms from the union of the anterior tibial, posterior tibial, and fibular veins; becomes the **femoral vein** after leaving the popliteal fossa.
- Immediately before entering the abdominal wall, the femoral vein receives: (1) the great saphenous vein, (2) the **deep femoral vein** (drains deeper thigh structures), (3) the **femoral circumflex vein** (drains the area around the femoral neck/head). The femoral vein then penetrates the body wall and **enters the pelvic cavity as the external iliac vein**.
- **Veins Draining the Pelvis:**
 - **External iliac veins:** receive blood from the lower limbs, pelvis, and lower abdomen; travel across the inner ilium surface.
 - **Internal iliac veins:** drain the organs of the pelvis; formed by the union of the **gluteal, internal pudendal, obturator, and lateral sacral veins**.
 - The **external iliac vein** merges with the **internal iliac vein** to form the **common iliac vein**.
 - The **median sacral vein** (drains the region supplied by the median sacral artery) empties into the **left common iliac vein**.
 - The left and right common iliac veins ascend obliquely and unite anterior to vertebra **L5** to form the **inferior vena cava**.
- **Inferior vena cava (abdominal portion):** retroperitoneal, ascends parallel and to the right of the aorta; collects blood from six major abdominal veins: (1) **lumbar veins** (connect to azygos/hemi-azygos veins); (2) **gonadal veins** (right gonadal vein → IVC directly; left gonadal vein → left renal vein); (3) **hepatic veins** (from the liver, enter IVC at vertebra T8 level); (4) **renal veins** (largest vessels draining into the IVC); (5) **adrenal veins** (right → IVC directly; left → left renal vein); (6) **phrenic veins** (right → IVC directly; left → left renal vein). Blood from the IVC flows into the right atrium, mixing with SVC blood, then to the right ventricle and pulmonary circuit.
- **Hepatic Portal System** (context, not strictly pelvic but part of the abdominal venous network described here): begins in digestive organ capillaries, ends in liver sinusoids; the liver is the only digestive organ draining directly into the IVC. Blood from celiac/superior mesenteric/inferior mesenteric artery capillary beds flows into hepatic portal system veins rather than directly to the IVC. **Hepatic portal vein** forms from the fusion of the **superior mesenteric vein** and **splenic vein** (also receives inferior mesenteric vein via the splenic vein, plus gastric and cystic veins near the liver). Portal blood has unusually high glucose/amino acid content (absorbed from the gut) — delivered to the liver for storage, metabolic conversion, or excretion, keeping systemic blood composition stable regardless of digestive activity.

22.7 | Cardiovascular Changes at Birth (Fetal Circulation — pelvis-relevant via umbilical vessels)

- Fetal vs. adult cardiovascular differences reflect different respiratory/nutritional sources: fetal lungs are collapsed/nonfunctional, the digestive tract has nothing to digest; the **placenta** provides all fetal respiratory and nutritional needs via maternal-fetal exchange.
- **Two umbilical arteries** branch off the **internal iliac arteries** of the fetus, enter the umbilical cord, and deliver (deoxygenated) blood to the placenta. Blood returns to the fetus from the placenta via the single **umbilical vein**, which carries oxygen and nutrients. The umbilical vein drains into the **ductus venosus**, connected to a venous network bypassing the liver.

- At birth, when the placental connection is severed, blood flow ceases in the umbilical vessels and they degenerate.
- **Fetal cardiac shunts:** the interatrial septum is functionally incomplete before birth — the **foramen ovale** (interatrial opening) has a valve-like flap allowing blood to flow right atrium → left atrium (but not backward), letting blood bypass the pulmonary circuit. A second shunt, the **ductus arteriosus**, is a short muscular vessel connecting the pulmonary trunk and aortic trunk. With fetal lungs collapsed and pulmonary capillaries compressed, little blood flows through the lungs: during diastole, ~**25%** of blood arriving at the right atrium passes through the foramen ovale into the left atrium (bypassing the lungs), and **>90%** of blood leaving the right ventricle passes through the ductus arteriosus into the systemic circuit rather than continuing to the lungs.
- **At birth:** the infant's first breath expands the lungs and pulmonary vessels; smooth muscle in the ductus arteriosus contracts, isolating the pulmonary and aortic trunks and allowing blood to flow through the pulmonary circuit; rising left atrial pressure closes the foramen ovale's valvular flap, completing cardiovascular remodeling. In adults, the interatrial septum bears a shallow depression, the **fossa ovalis**, marking the former foramen ovale site; the ductus arteriosus remnant becomes a fibrous cord, the **ligamentum arteriosum**.
- **Clinical — Congenital cardiovascular problems:** if proper circulatory changes don't occur at/after birth, problems develop, severity depending on which fetal connection remains open and the opening's size; treatment may involve surgical closure of the foramen ovale, ductus arteriosus, or both. Other congenital heart defects result from abnormal cardiac development or inappropriate connections between the heart and major vessels. Congenital cardiovascular problems serious enough to threaten homeostasis are relatively rare, usually reflect abnormal heart formation or heart–great vessel interconnection problems; most are surgically correctable, though multiple surgeries may be needed and life expectancy may be shortened in severe defects.

22.8 | Aging and the Cardiovascular System

- **Key Point:** age-related changes occur in blood composition and in heart/blood vessel anatomy and function; cardiovascular capabilities gradually decline with age.
- **Blood changes:** (1) decreased hematocrit; (2) constriction/blockage of peripheral veins by a **thrombus** (stationary clot), which can detach, travel through the heart, and lodge in a small artery (commonly in the lungs) causing a **pulmonary embolism**; (3) pooling of blood in leg veins as venous valves work less efficiently.
- **Heart changes:** (1) reduced maximum cardiac output; (2) decreased elasticity of the cardiac skeleton; (3) progressive atherosclerosis restricting coronary circulation; (4) replacement of damaged cardiac muscle cells by scar tissue; (5) changes in pacemaker/conducting cell activity.
- **Blood vessel changes** (some related to arteriosclerosis): (1) decreased arterial/venous elasticity; (2) arterial walls becoming less tolerant of sudden pressure increases, which can lead to an **aneurysm** (enlarged blood vessel) — potentially causing stroke, heart attack, or massive blood loss depending on the vessel; (3) calcium salt deposition on weakened vascular walls, increasing stroke/heart attack risk; (4) thrombus formation at atherosclerotic plaques.